Hovercraft Competition

ASME PSU 2025

From the perspectives of Xiang

Preface

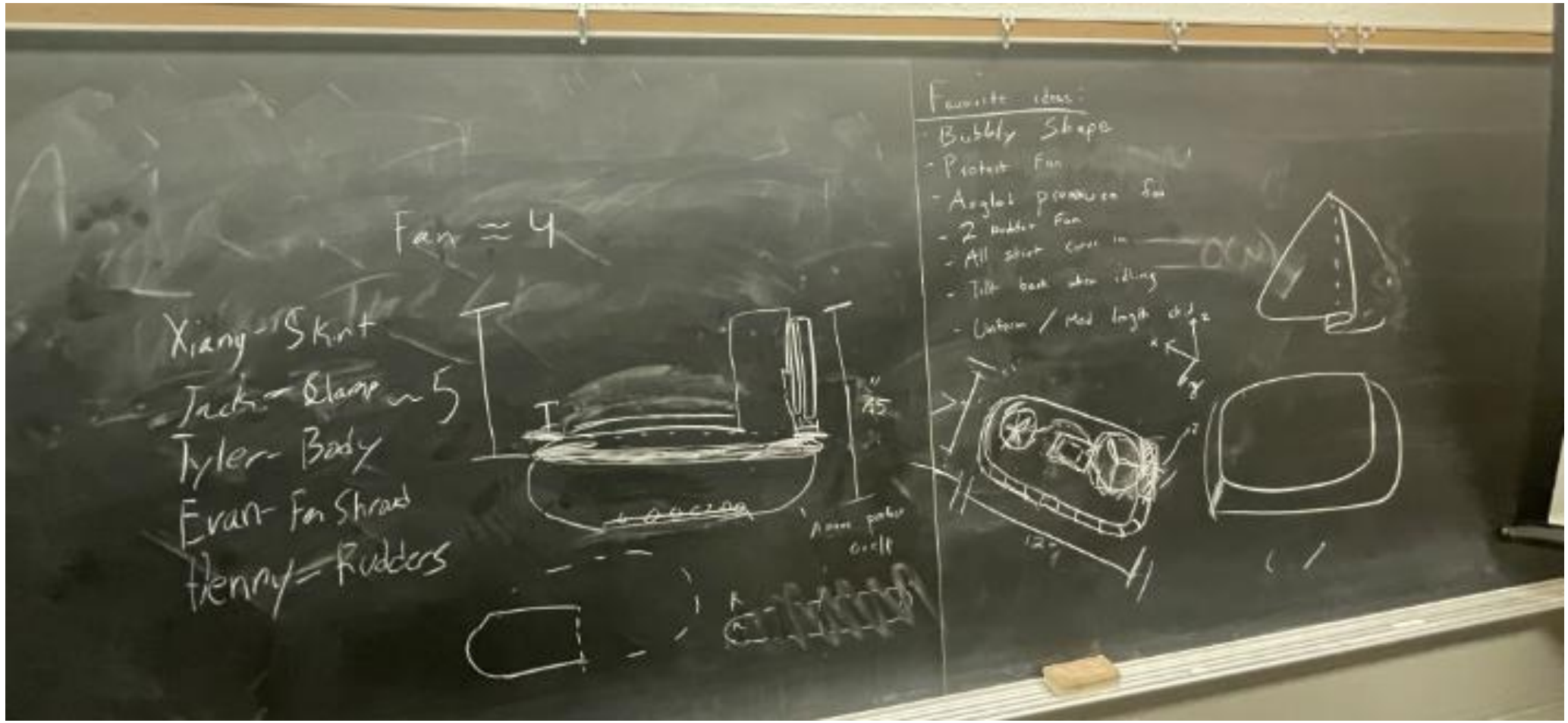
- This document is not necessarily formatted in any particular way.
- It contains:
  - Documented work
  - Explanations to others
  - General Progress

- It mostly contains my perspective on how the project progressed
- Some major time stamps have been noted for story purposes

Competition Detail

- Each team are given a standard set of parts
  - Motors, Batteries, ESCs
- Not allowed to buy stronger standard parts
- Can rebuy fried parts and will not count toward budget
- 100 USD
- 2 semesters
- No importing reference parts

Brainstorming Session

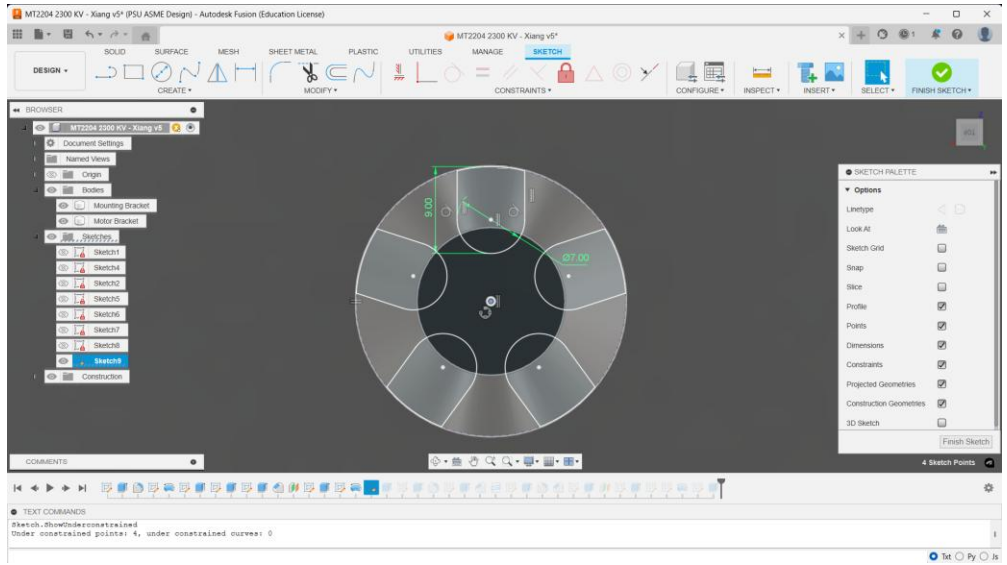


Brainstorming Session Summary

- The hover craft will have
- 1 Chassis
- 1 Propeller Fan
- 1 Lift fan
- 1 Skirt

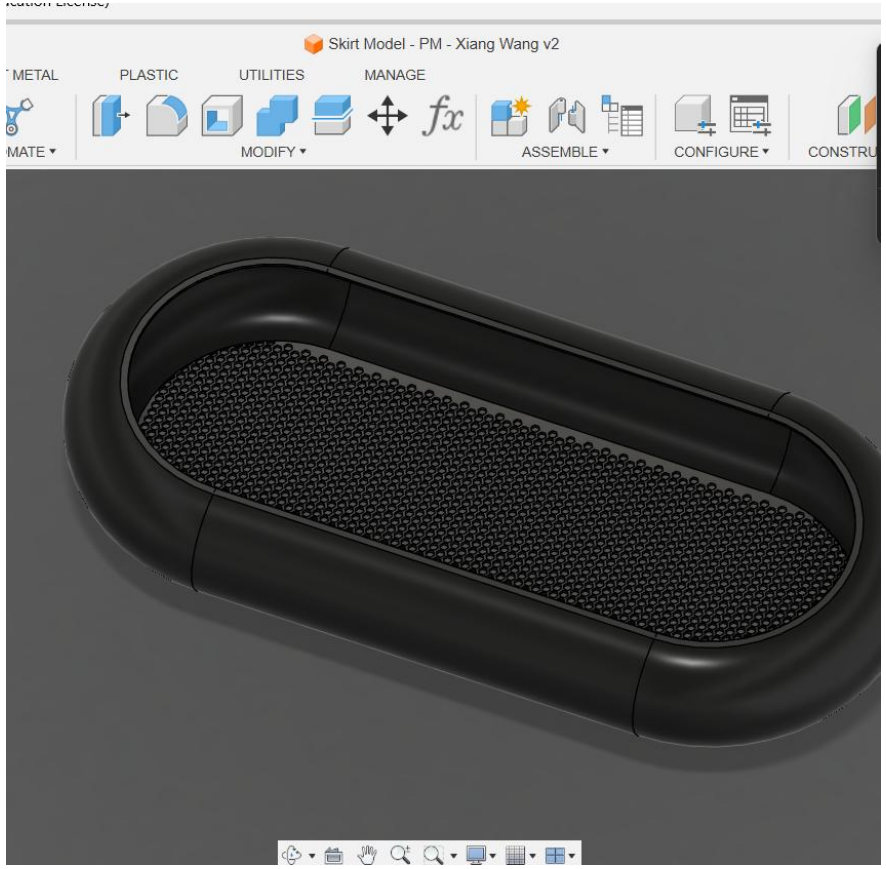
We like having a rudder, angled lift fan, and forward lift fan

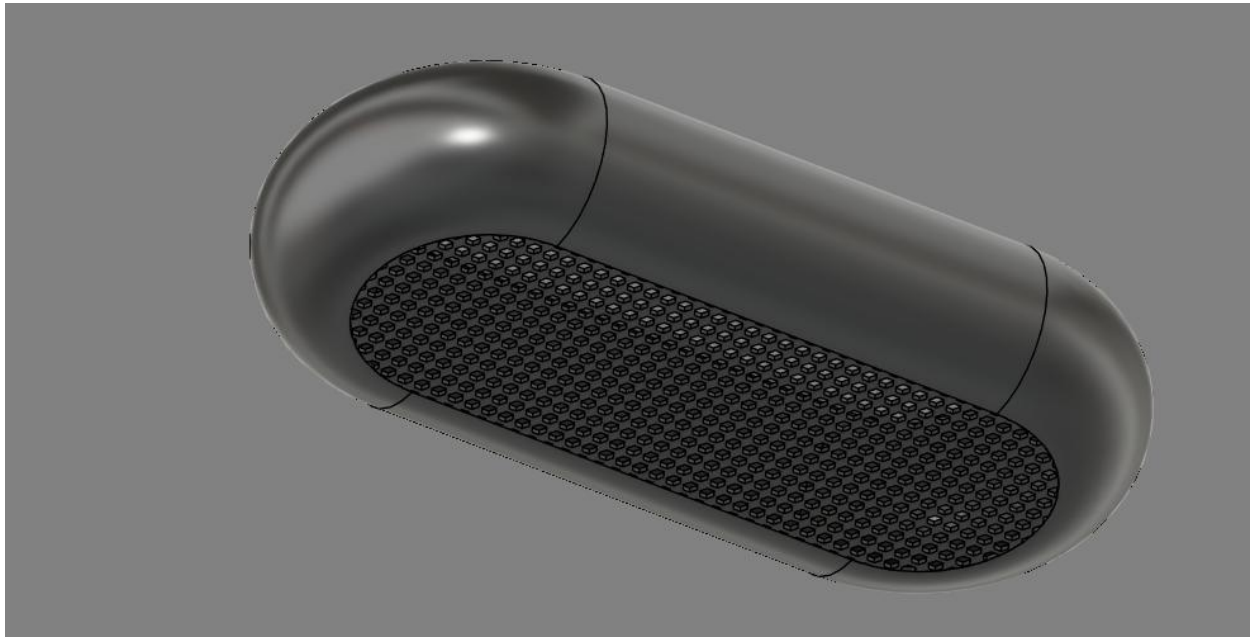
Making Reference Parts (MT2204 – 2300 KV)



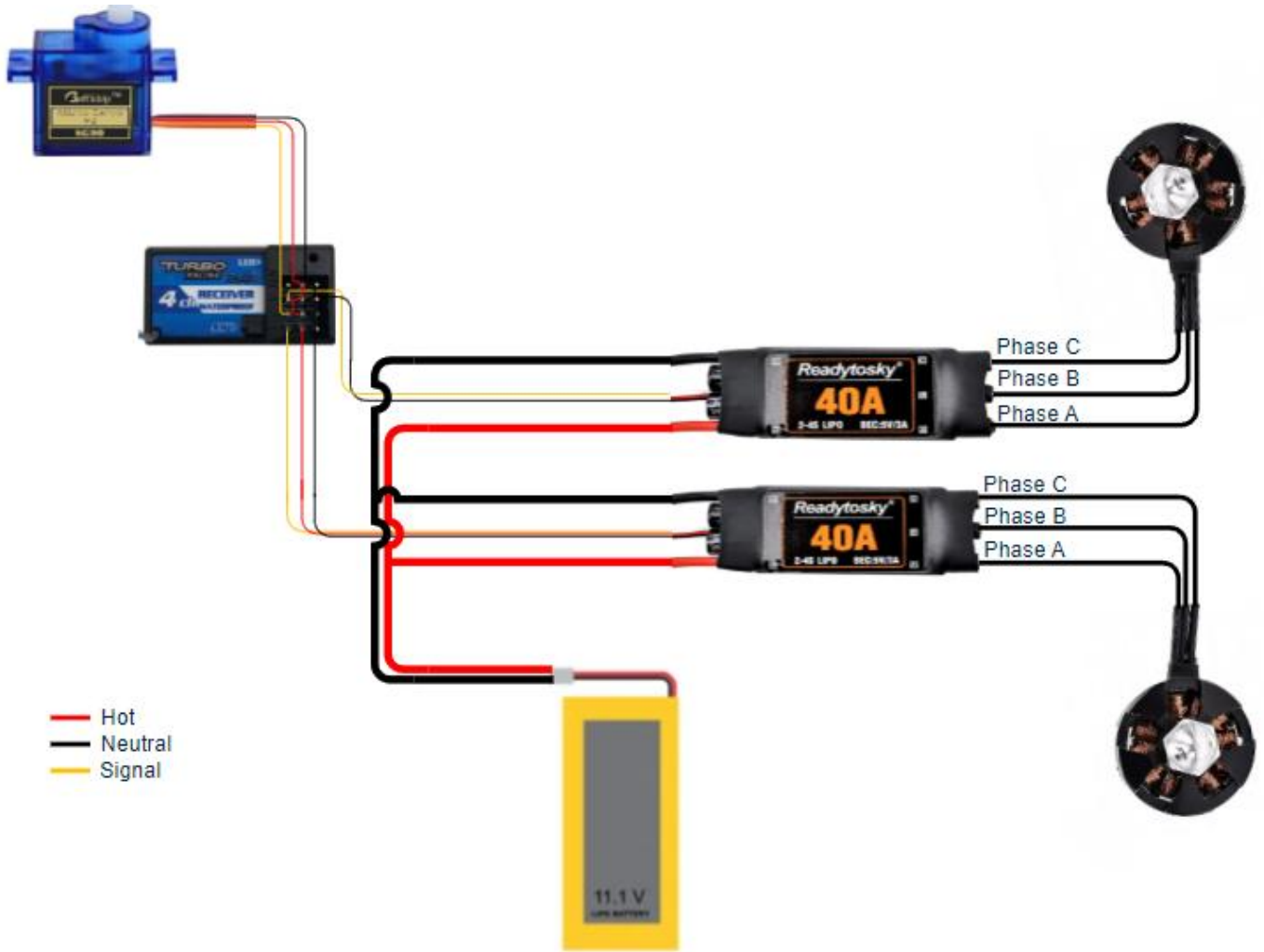


Making the Skirt Reference

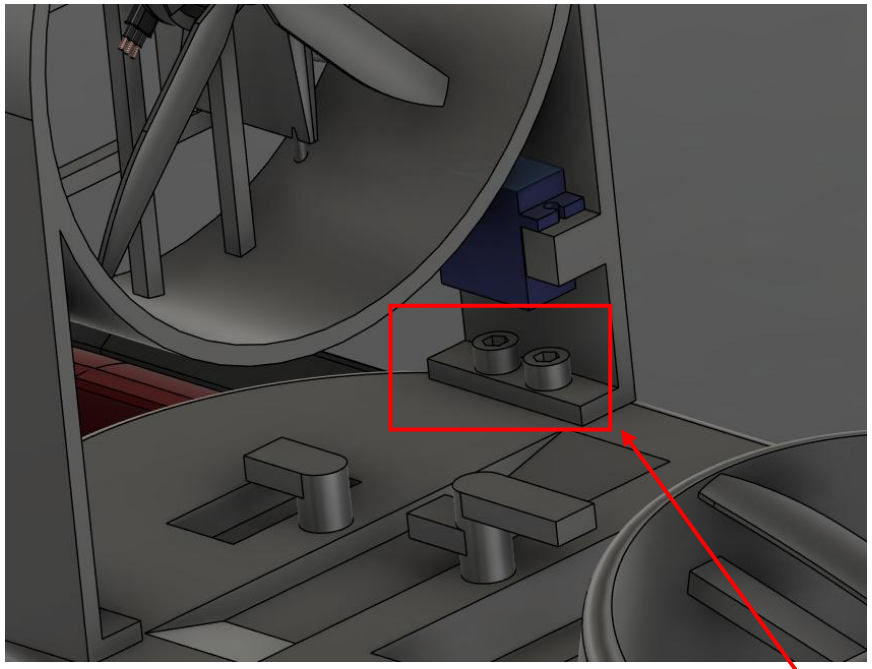


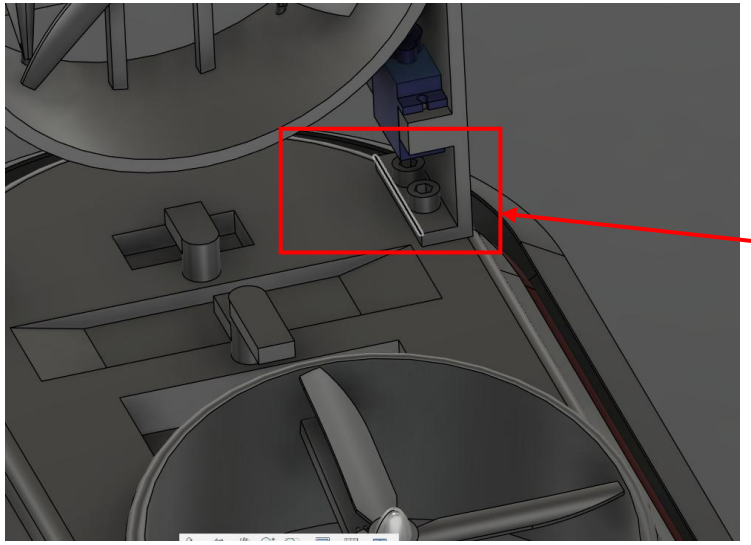


Wiring Diagram

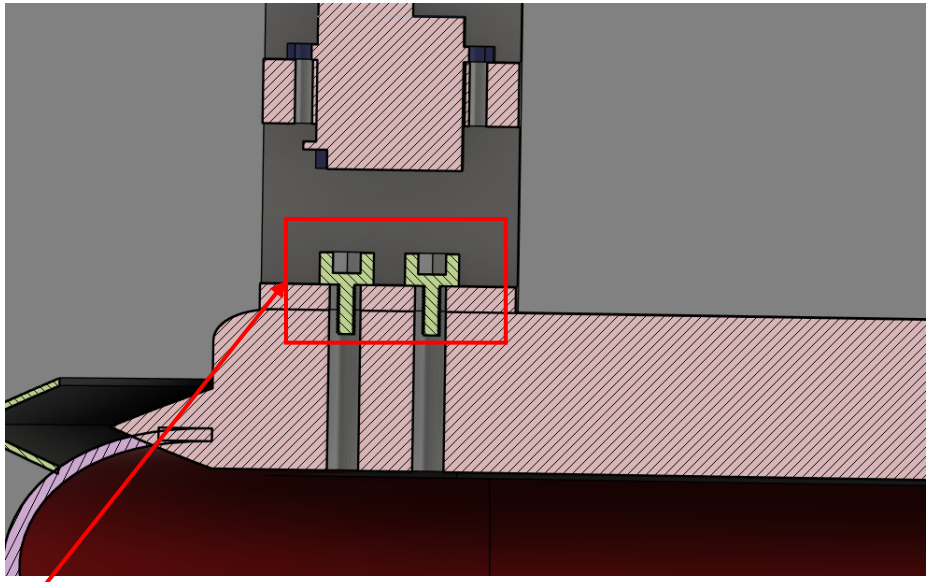


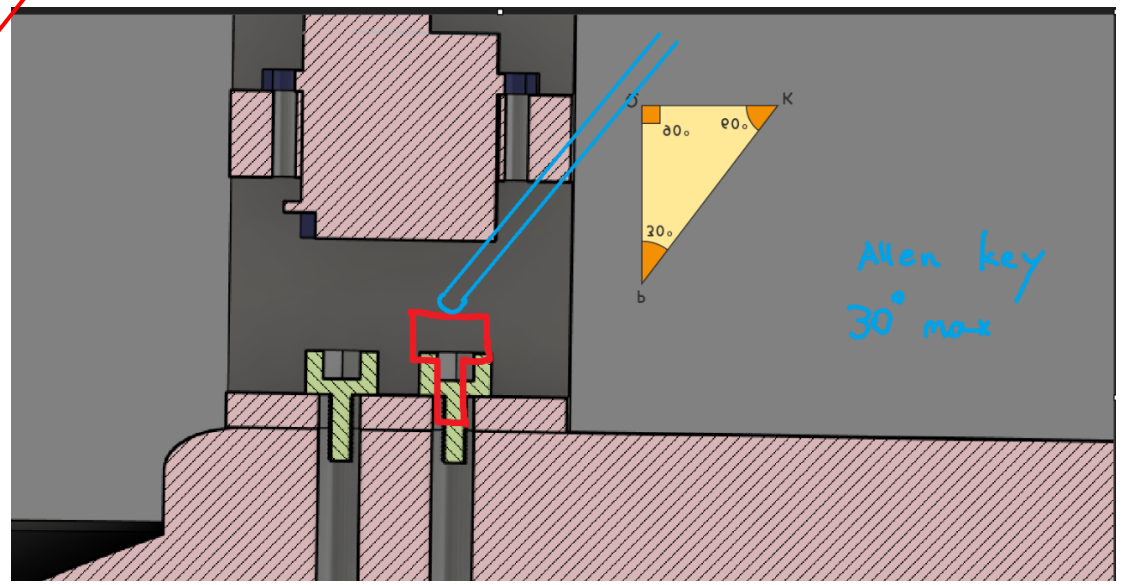
Does this work?



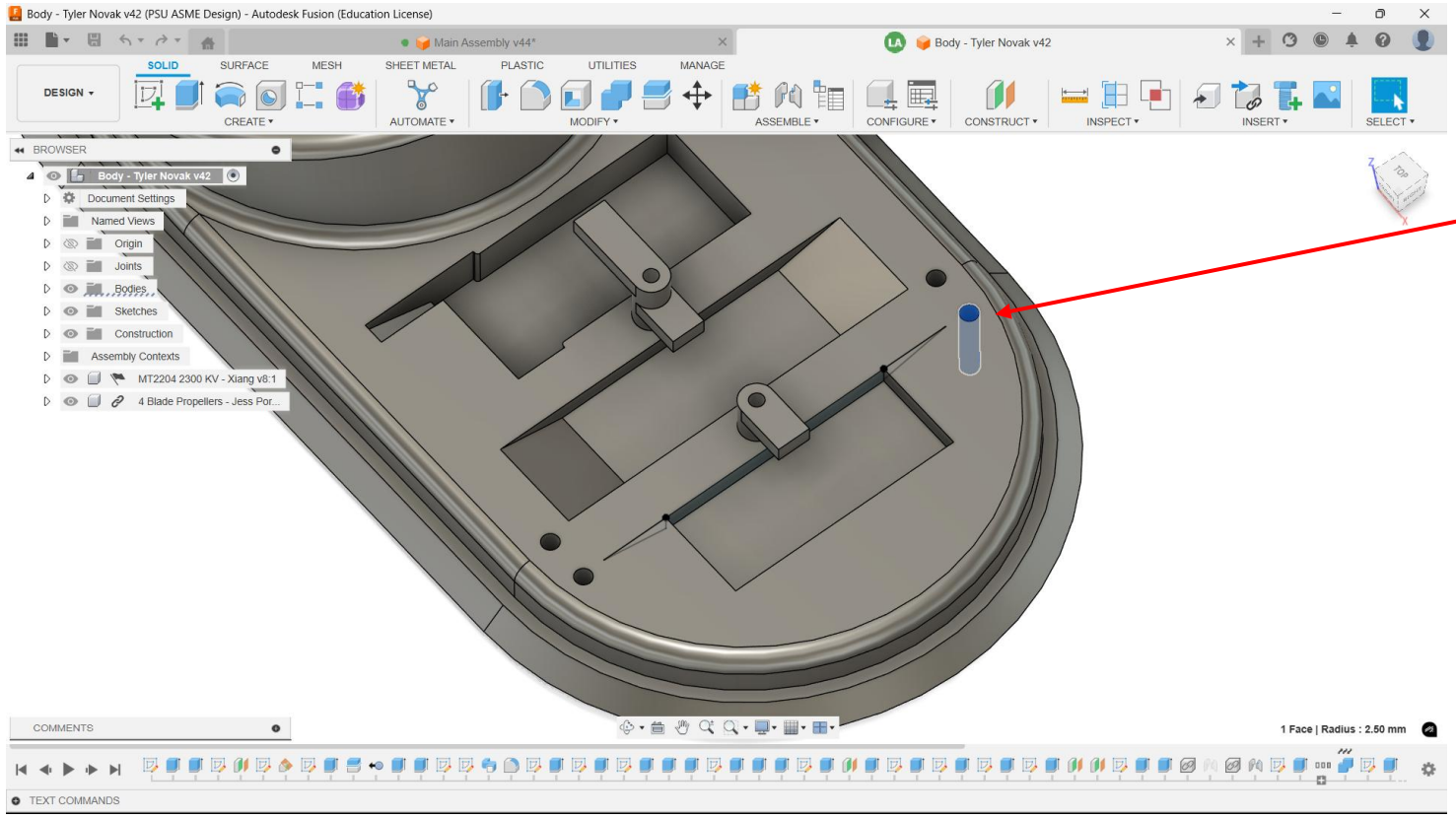


Hard to remove
In its current
position



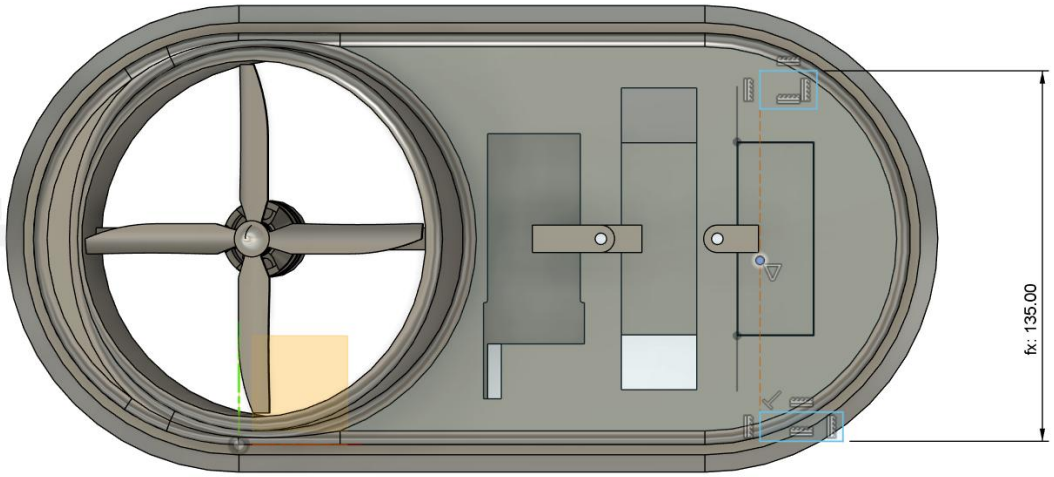


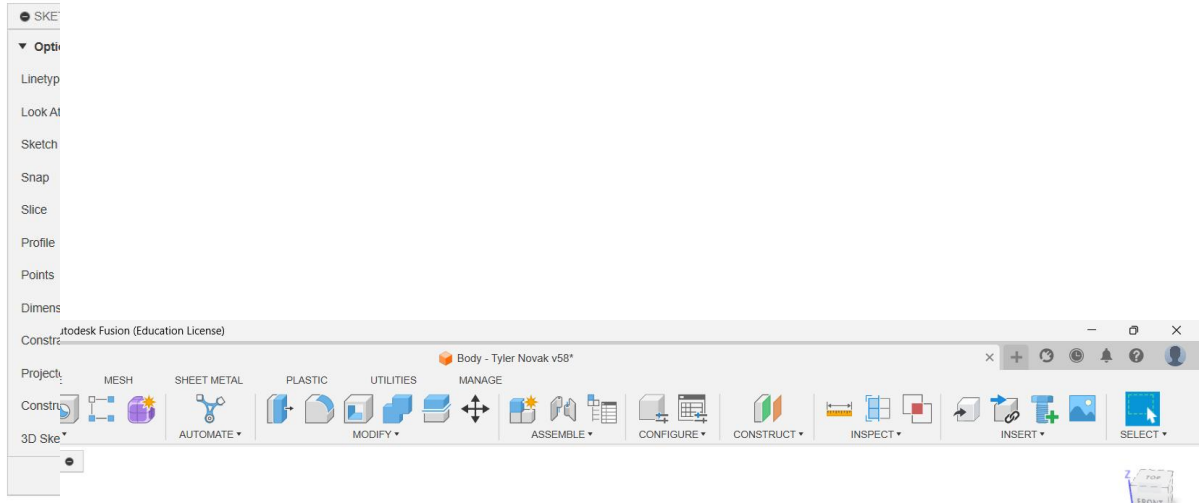
Let’s remove these



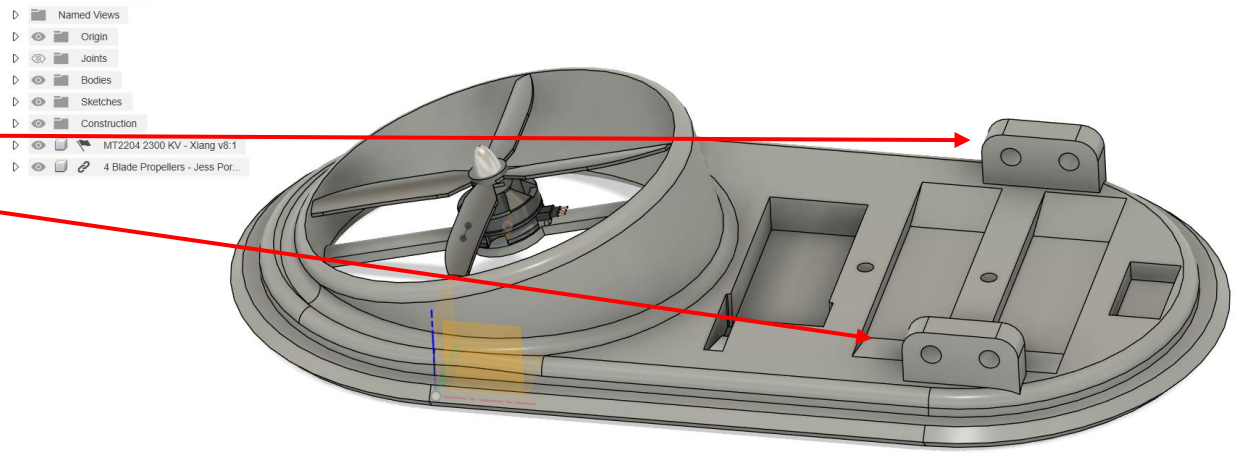
Let’s change these

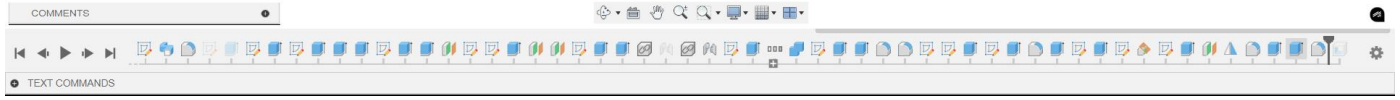
Raised Mounting Point Suggestion



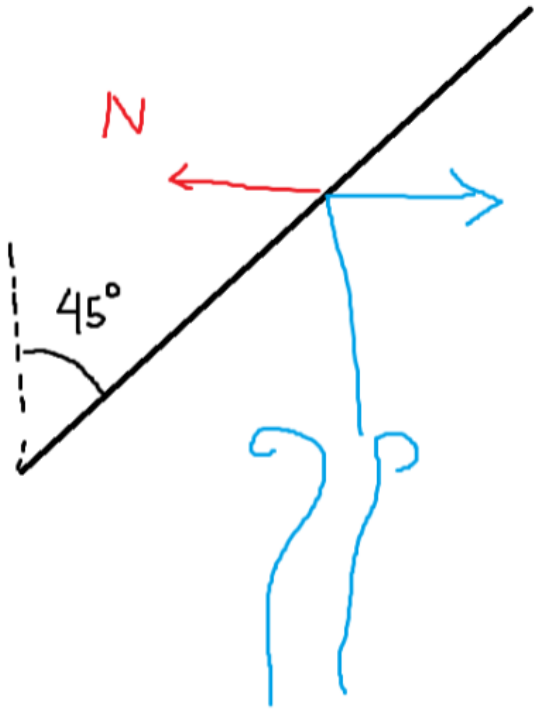


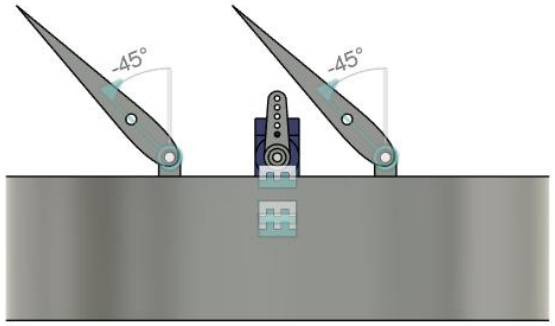
Easier to assemble





Maximum angle needs

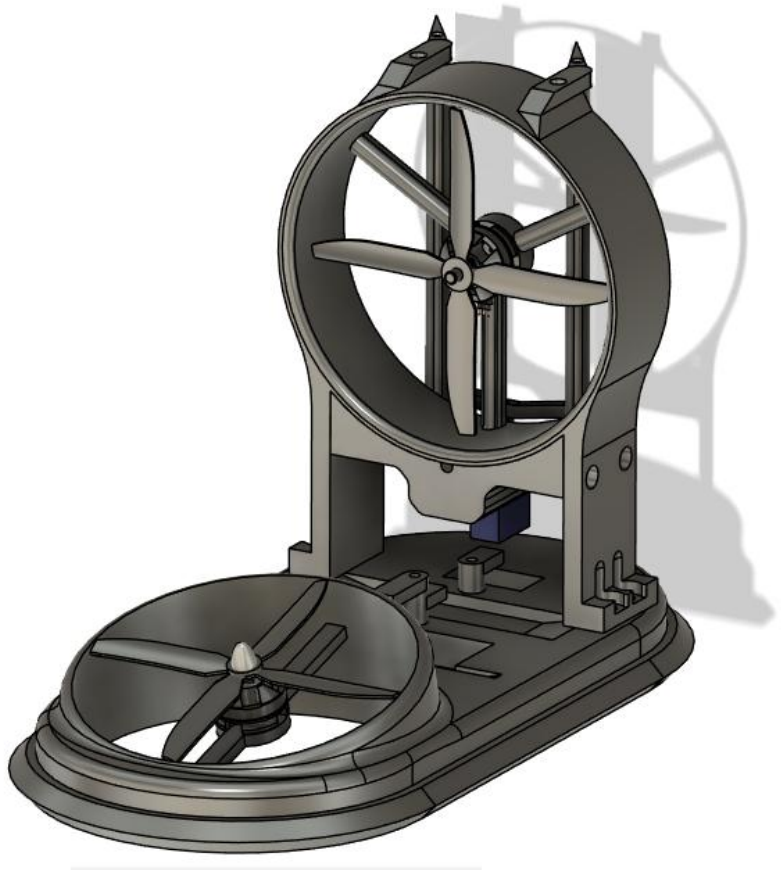




Max turning torque when these rudders are 45 degrees

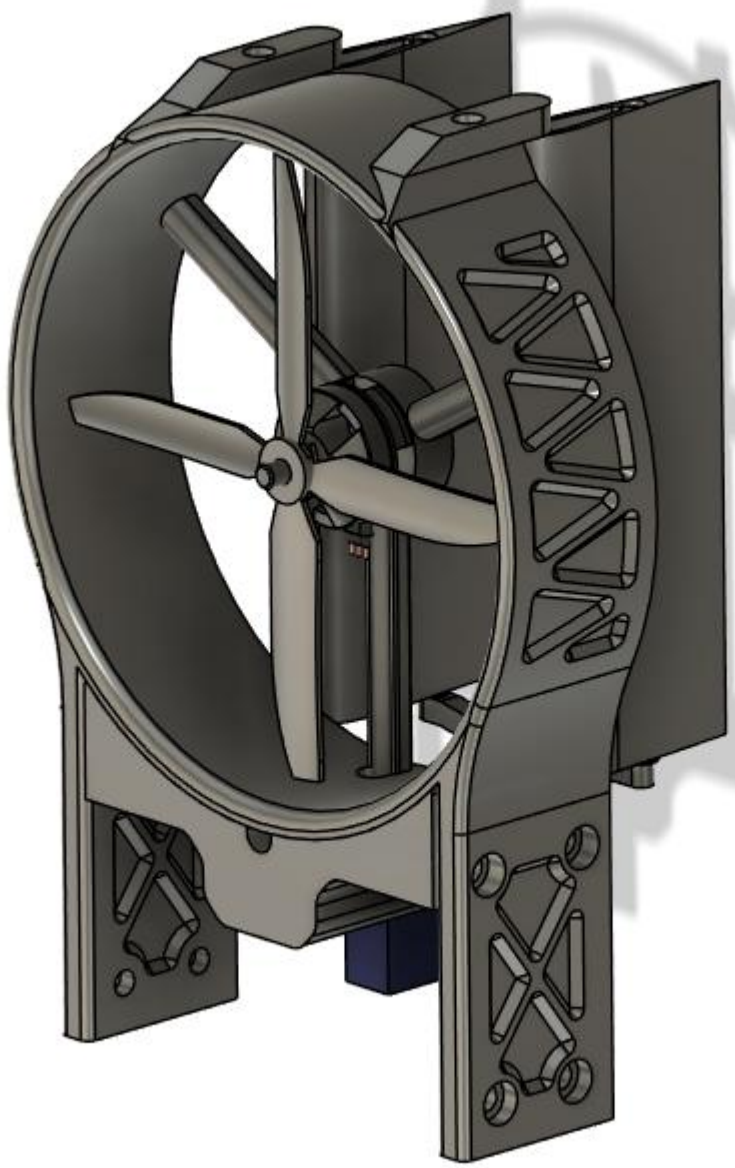
So we only need to make them go from -45 to 45 AT MOST

V1 Completion

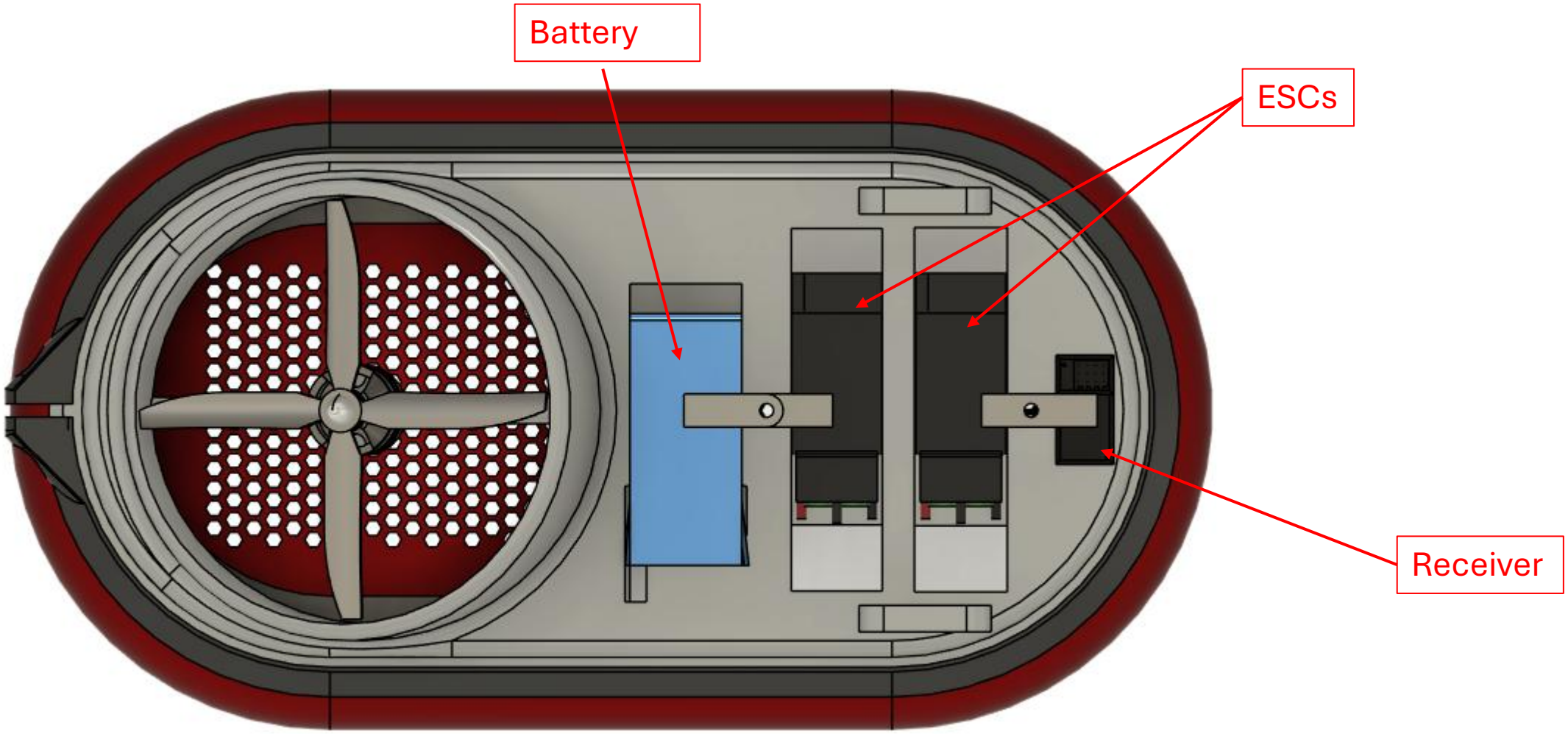


Isogrids Proposal?

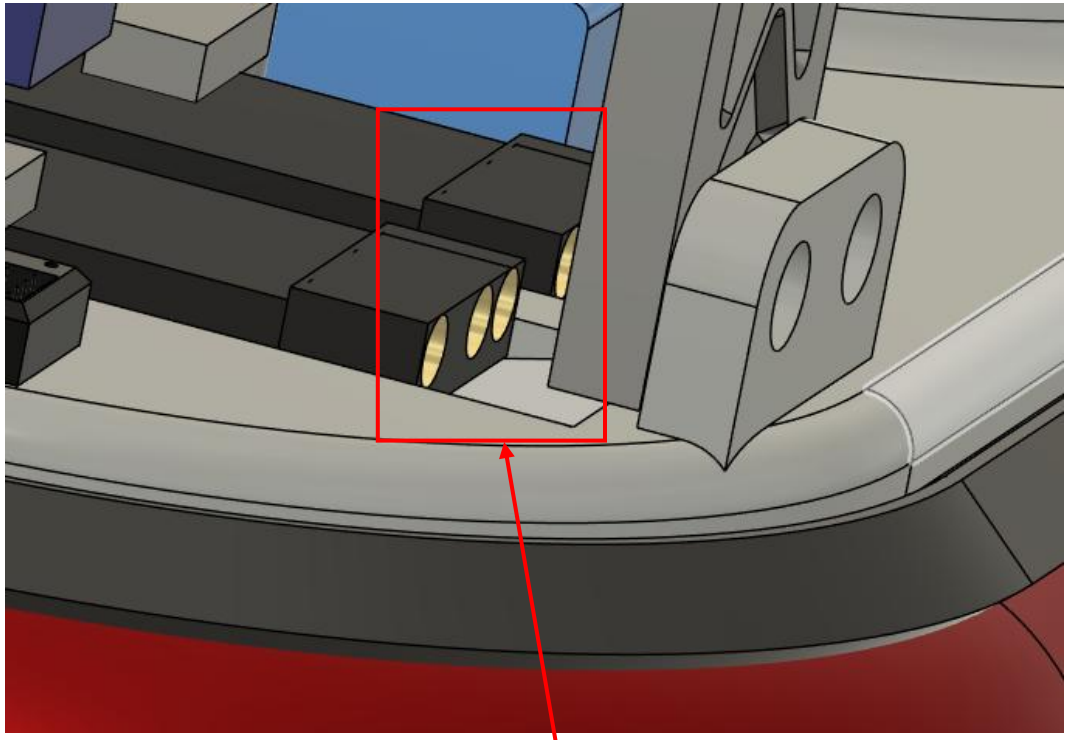
- May help with weight reduction
- Looks kind of cool



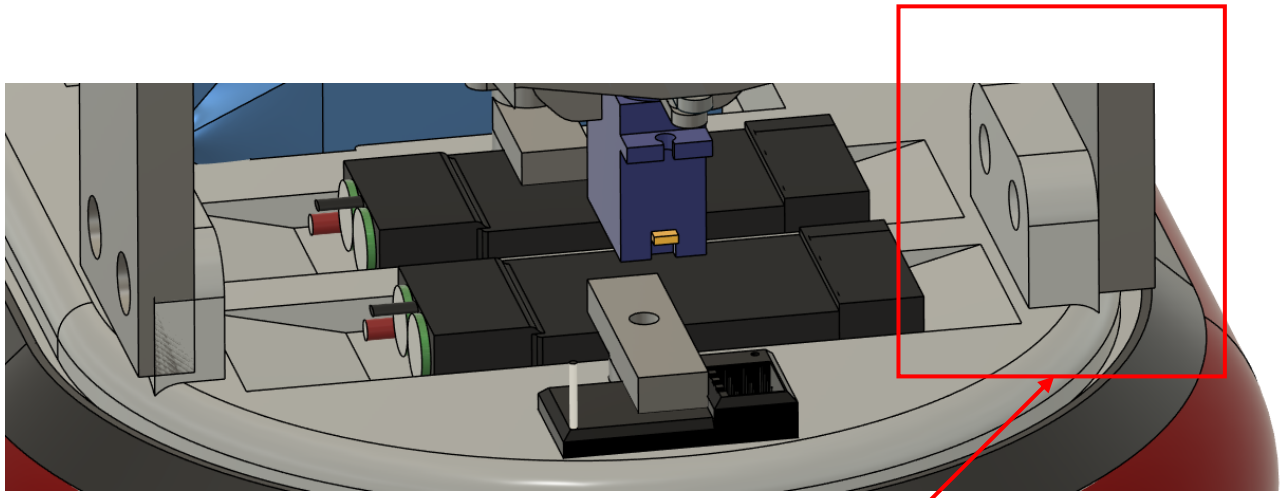
Current Component Layout



Component Interference



We need more space here, the ESC connectors are very big and will run into the mount



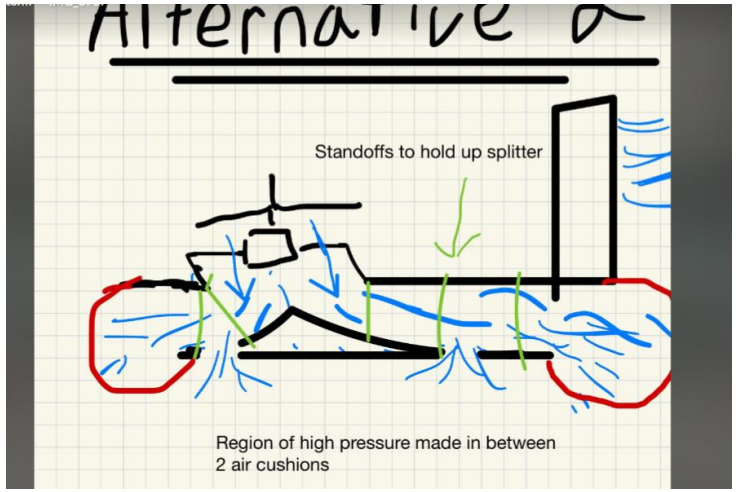
Let’s flip this to the outside so it doesn’t run into the ESCs

First flight!



First Flight Issues

Bobbing issue, maybe this might help

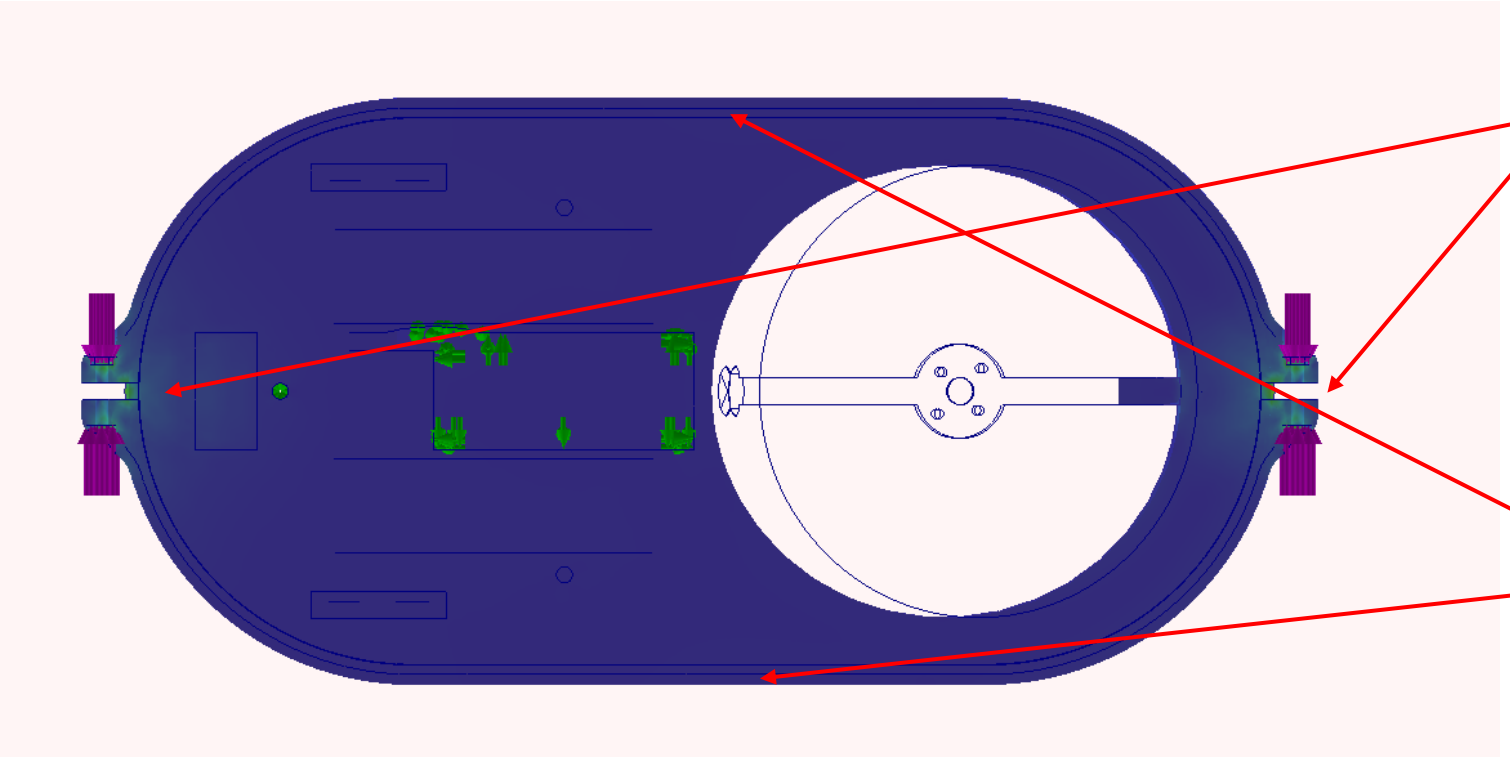


The propeller were pushing the hovercraft into the ground.

Angled lift fan was also kind of odd

The hovercraft skirt kept coming out

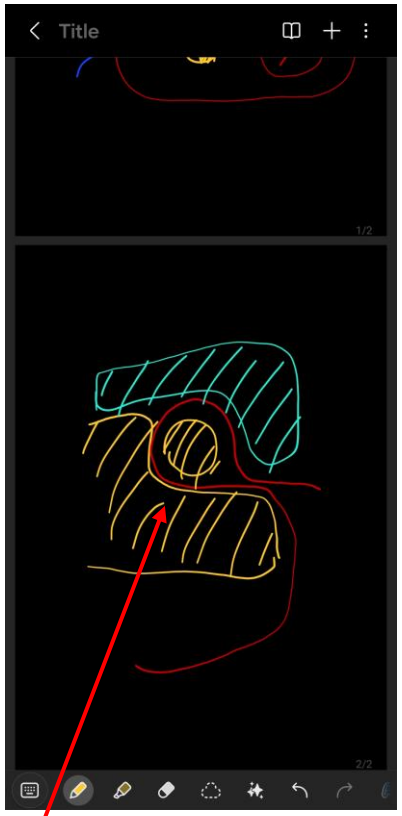
Clamping Force Issues



This reminds me of the problem I saw in this video from years ago lol https://youtu.be/1RJunnhpFIM

Clamping redesign force proposal





What if we did something like this

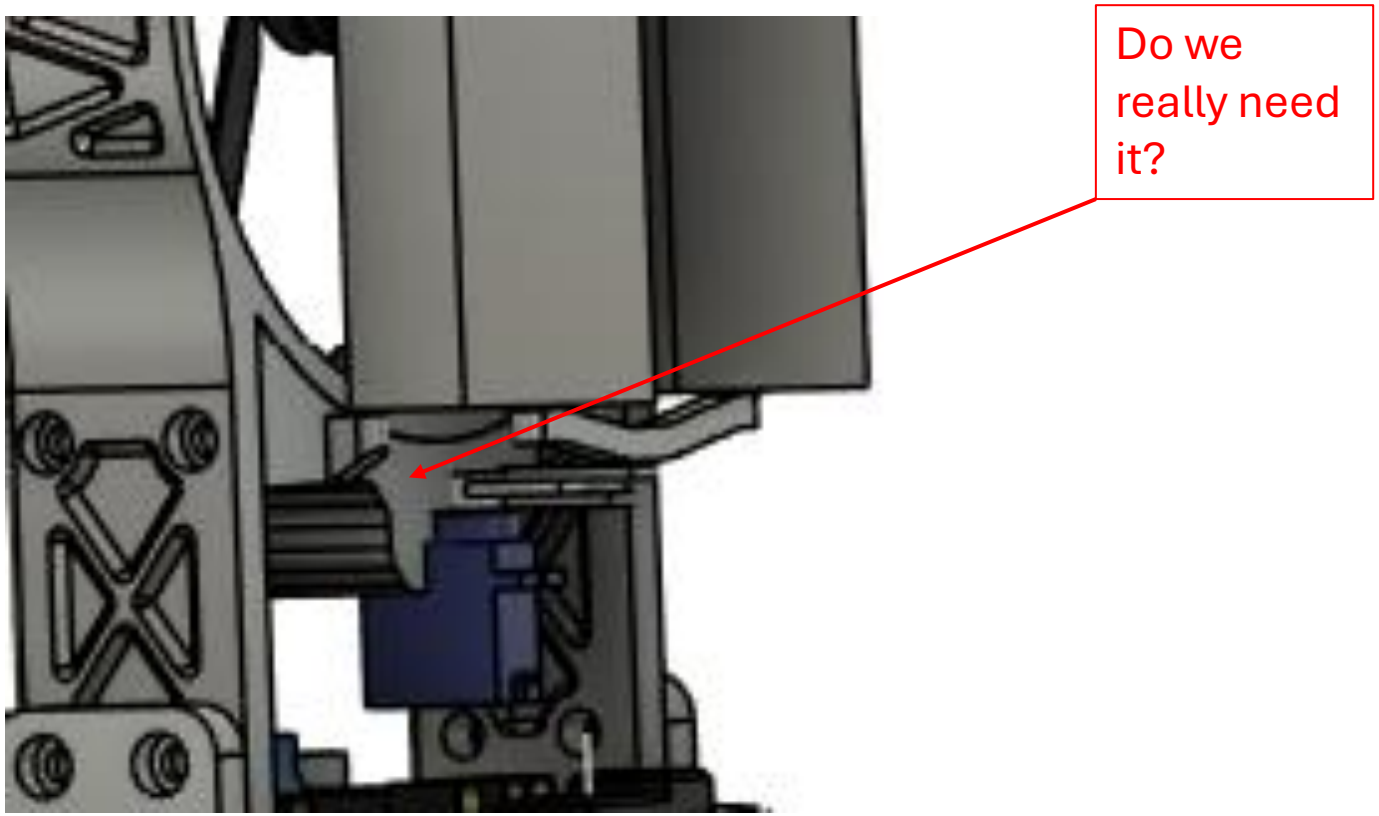
Red is skirt

Yellow is chassis

Cyan is new clamp

And the circle is some squishy material to press of it together

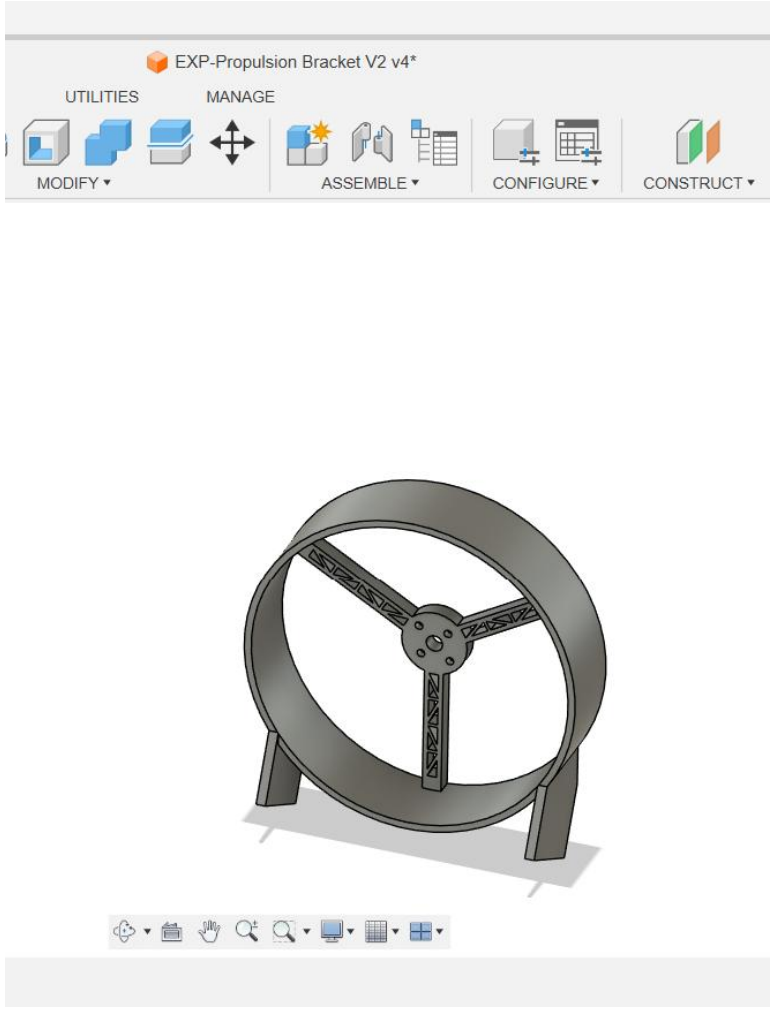
Part too bulky



Benefits:
- Reduced weight
- Simplified parts
- Less assembly

Work:
- Need to make sure the legs are still strong (FEA?)

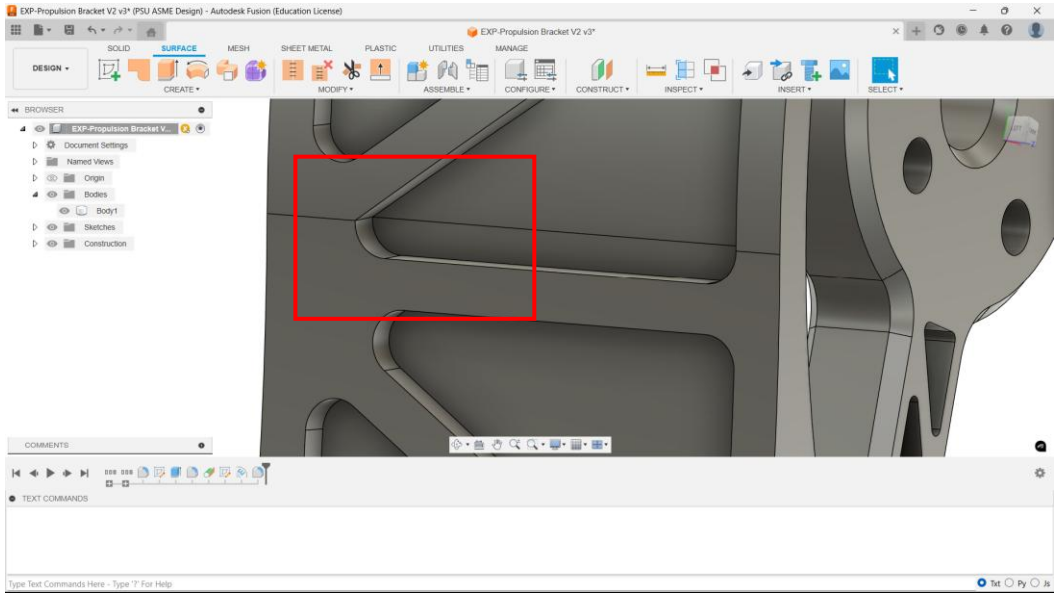
New Propulsion Bracket Design Proposal



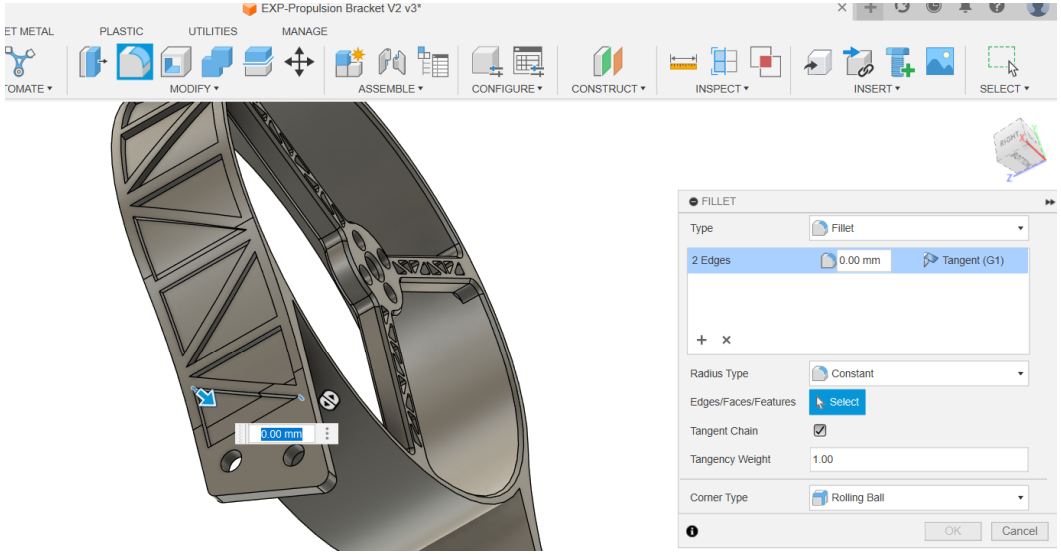
Redo

- Rudder Mount
- Servo Mount
- Leg Mount

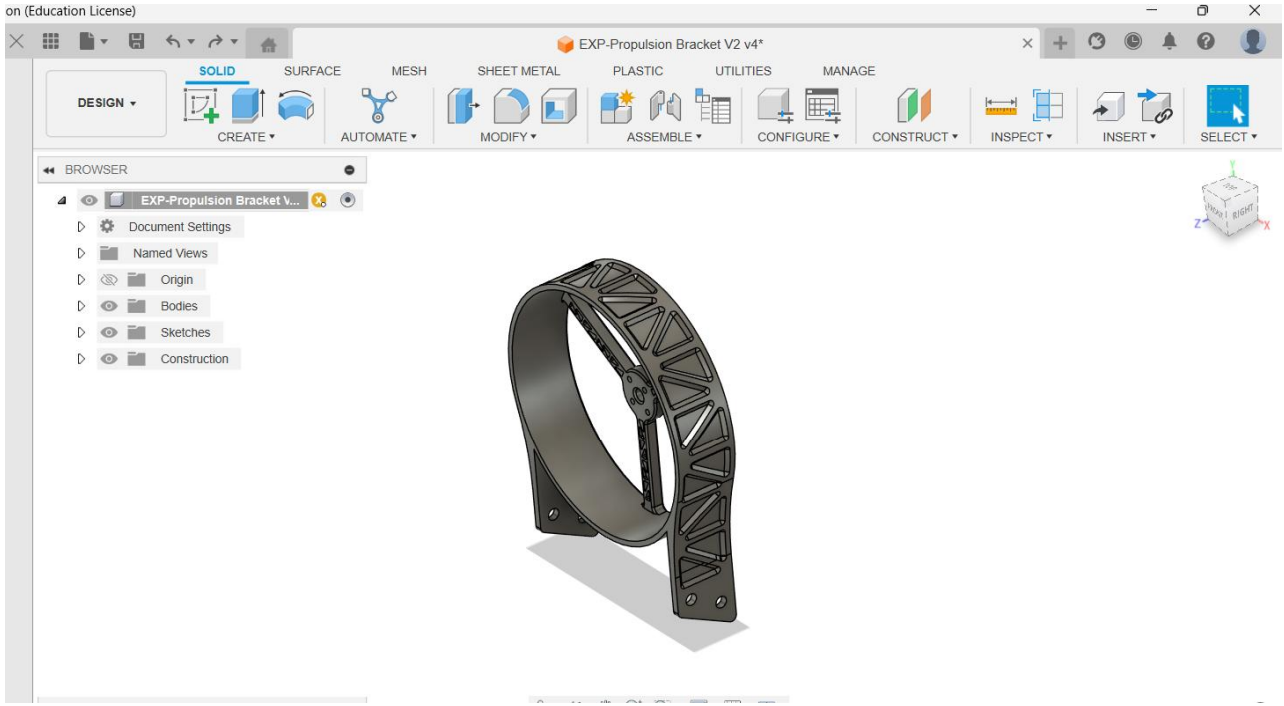
Isogrid Continuity Issue

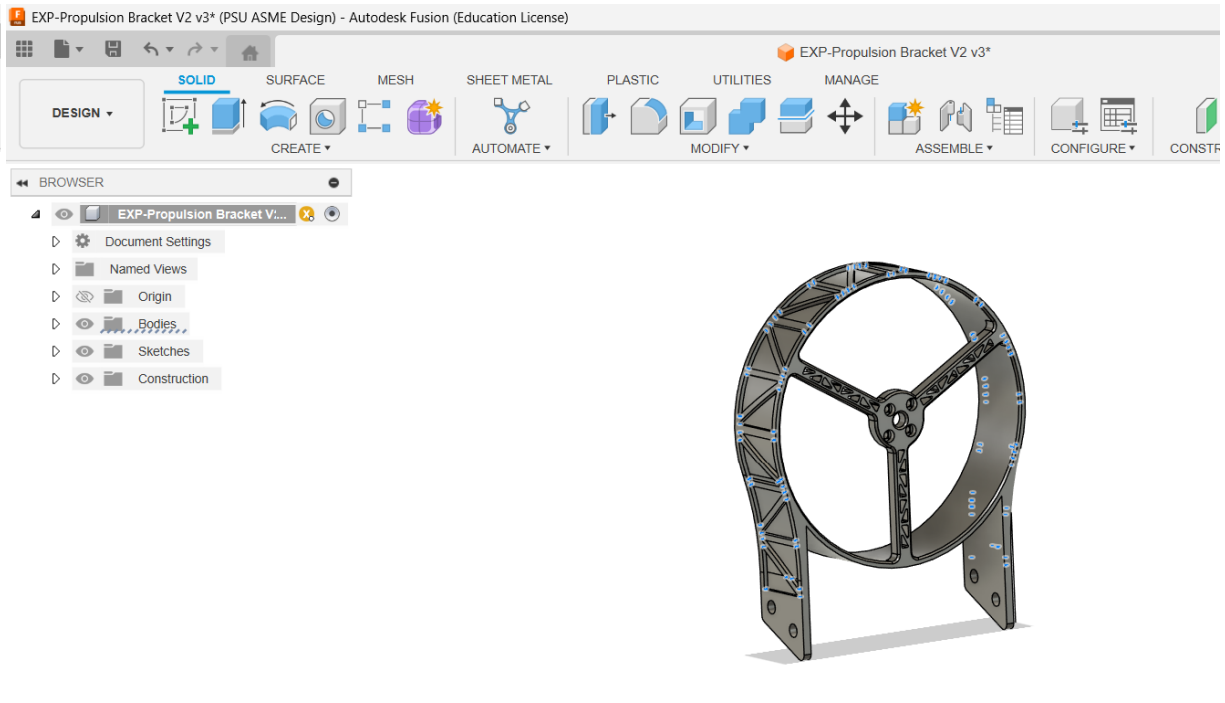


What if we did it like this

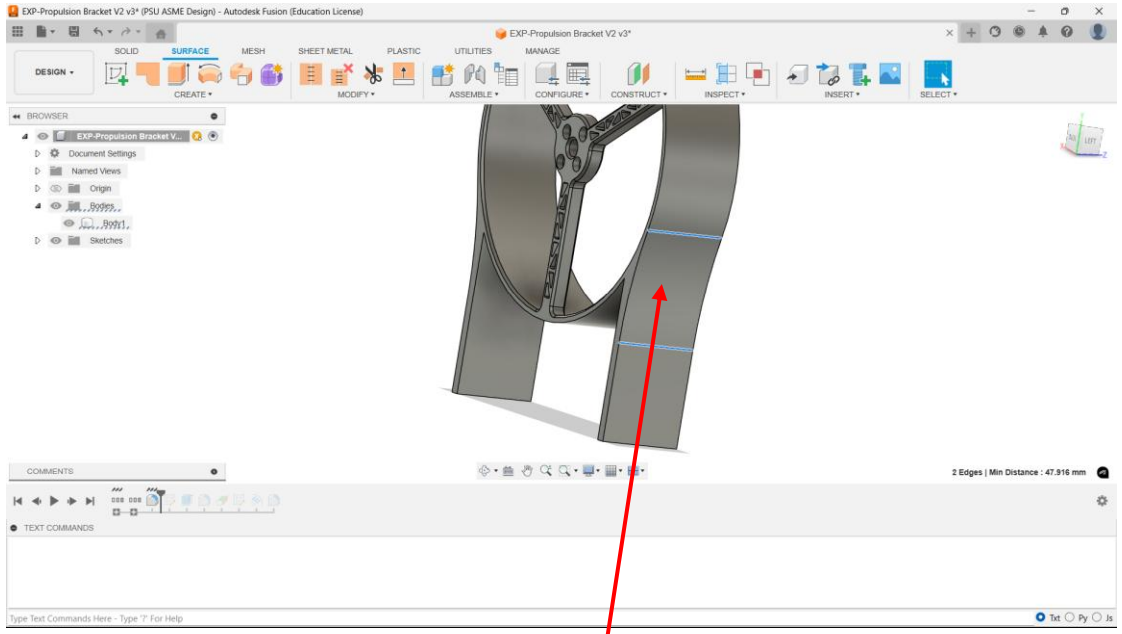


Isogrid Redesign



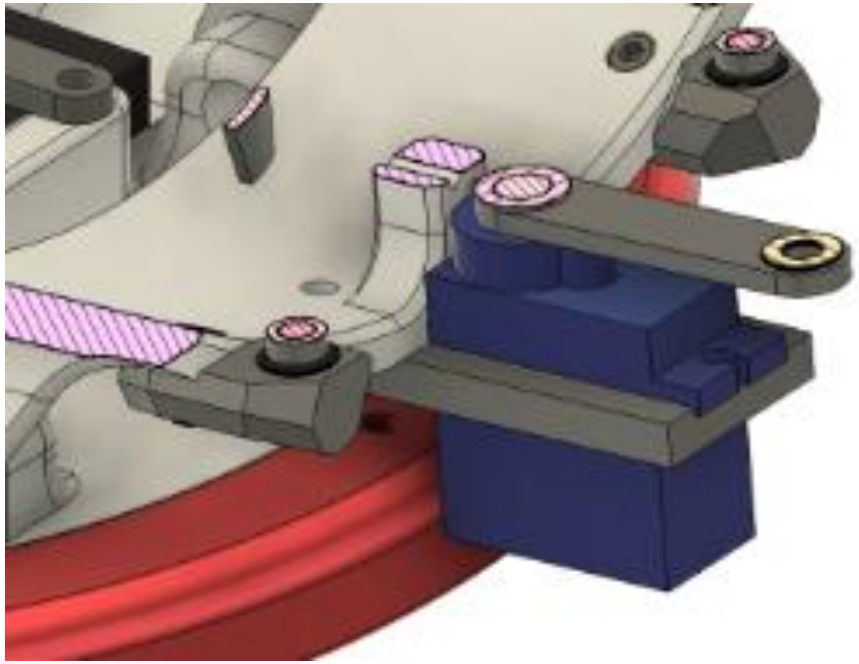


Isogrid Redesign



Nevermind isogrid can be fixed by increasing the radii here

Redesigning the servo horn



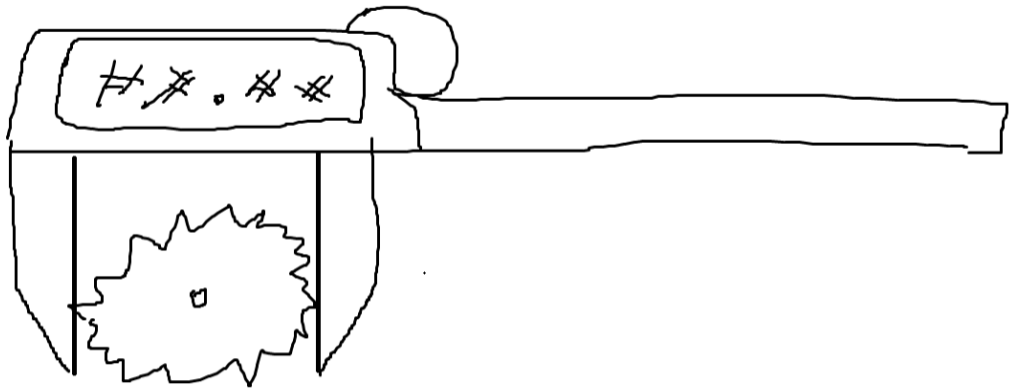
Benefits

- Much simpler. No more horn that was shipped with manufacturer.
- Reduced assembly
- Weighs less

Work

- We need to actually be able to manufacture it. No FDM printing; not enough resolution. Possible resin?

We need custom Servo Horn



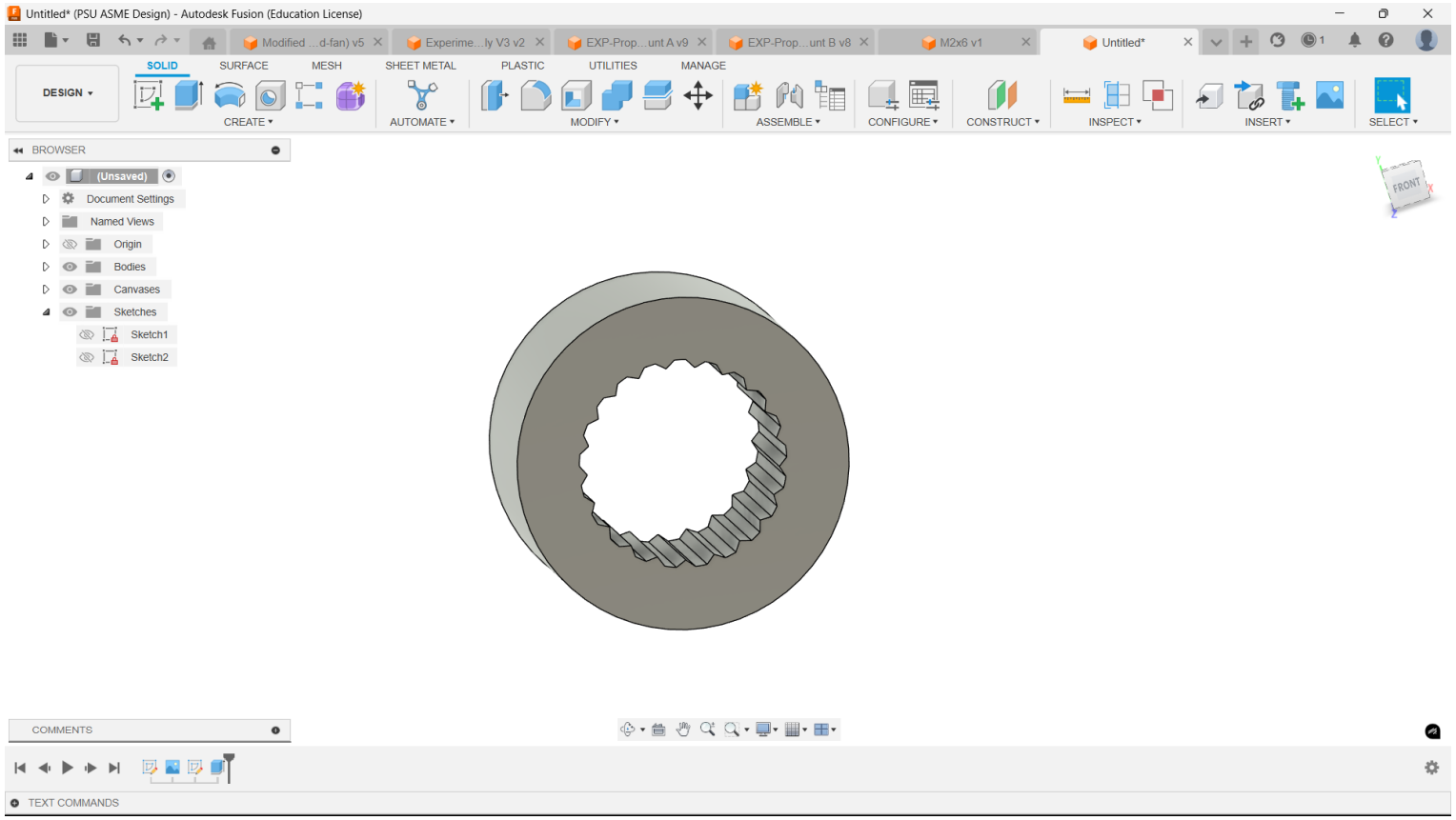


Really hard to measure, so what if you just did it like this.

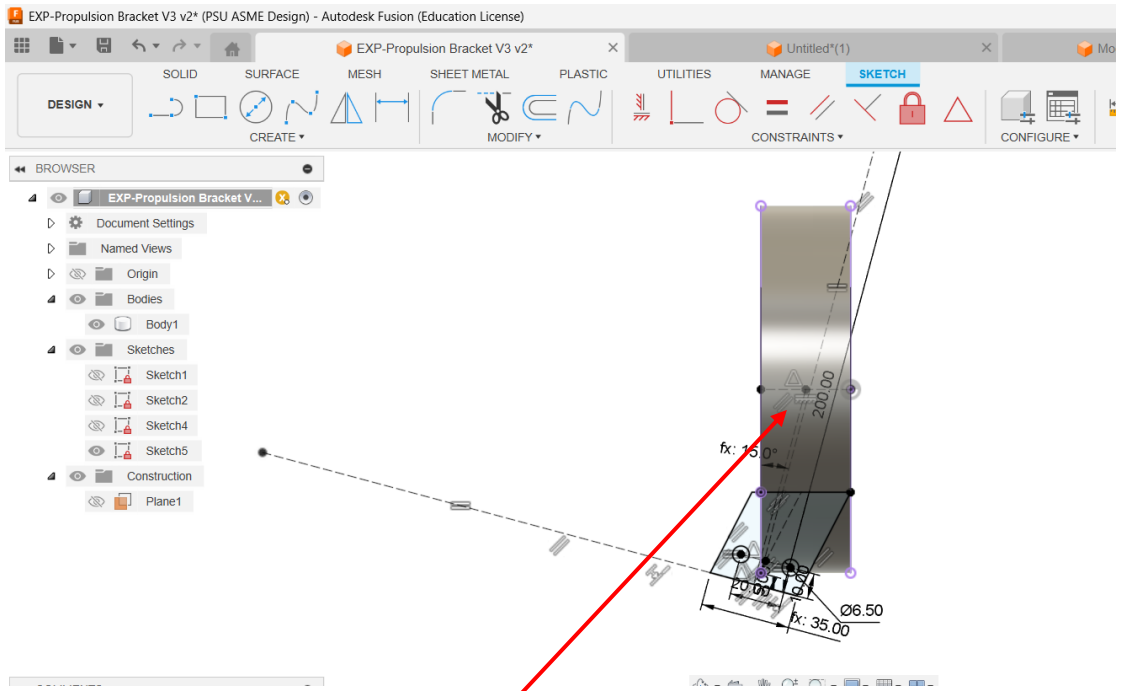
Get a standard reference like outer diameter. Then I can go and manually draw the teeth in CAD

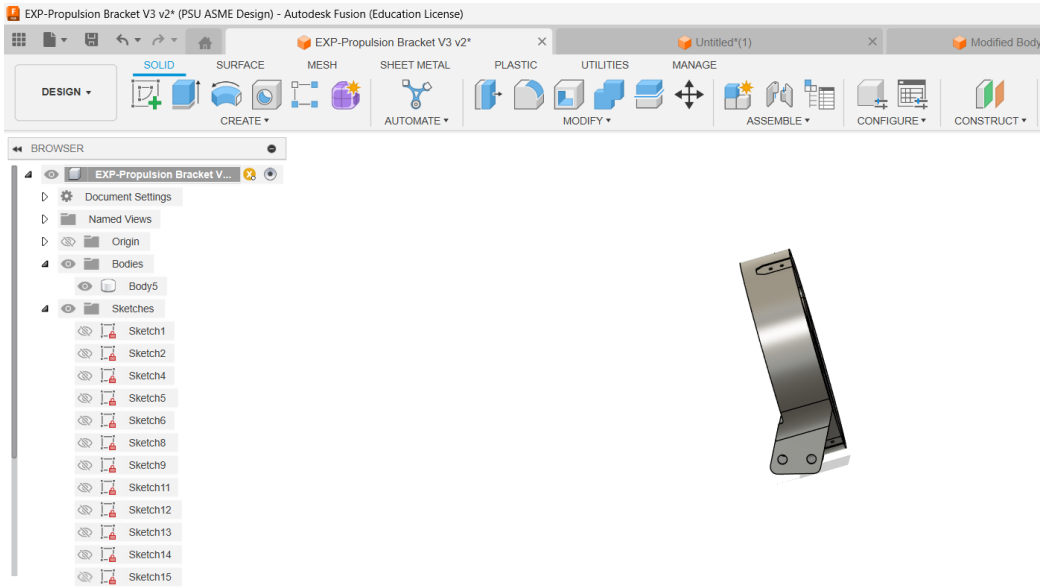
Thanks for measuring, George!

Servo Spline Profile



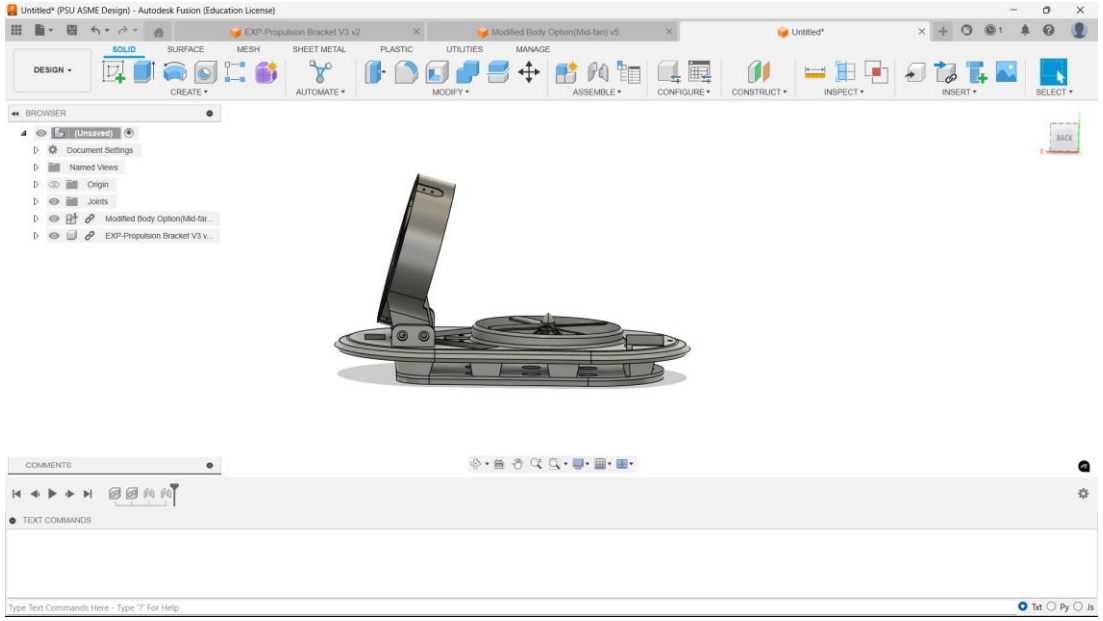
Propulsion Bracket Recommendations



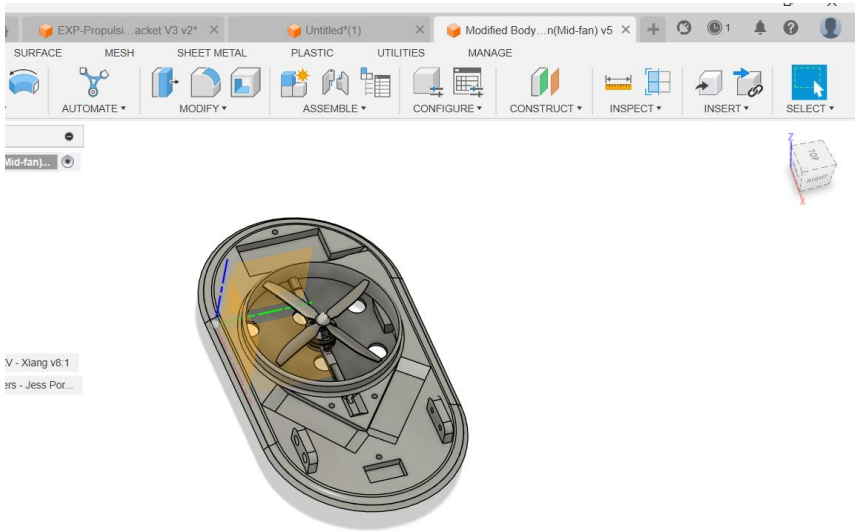


What if we angled the center of thrust right of the middle of mount. So forces are easier to calculate

V2 Changes



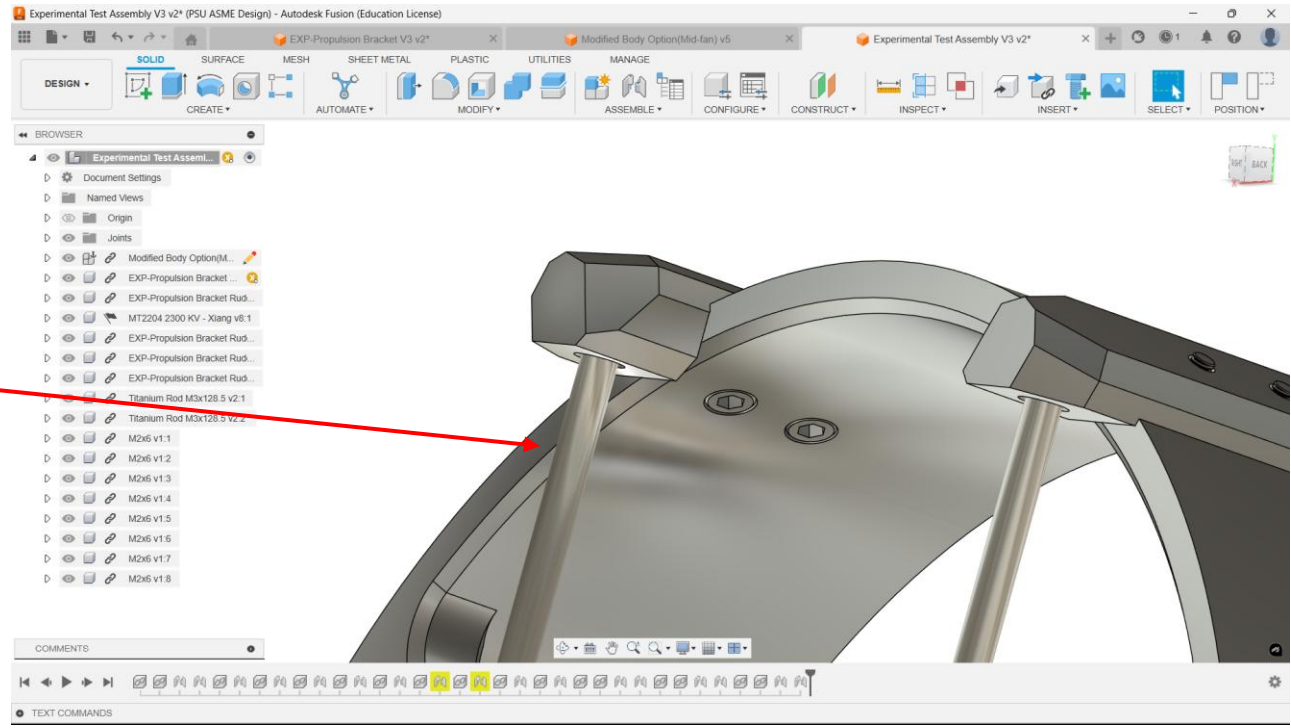
Flat thrust fan
Better part placement
Angled propeller

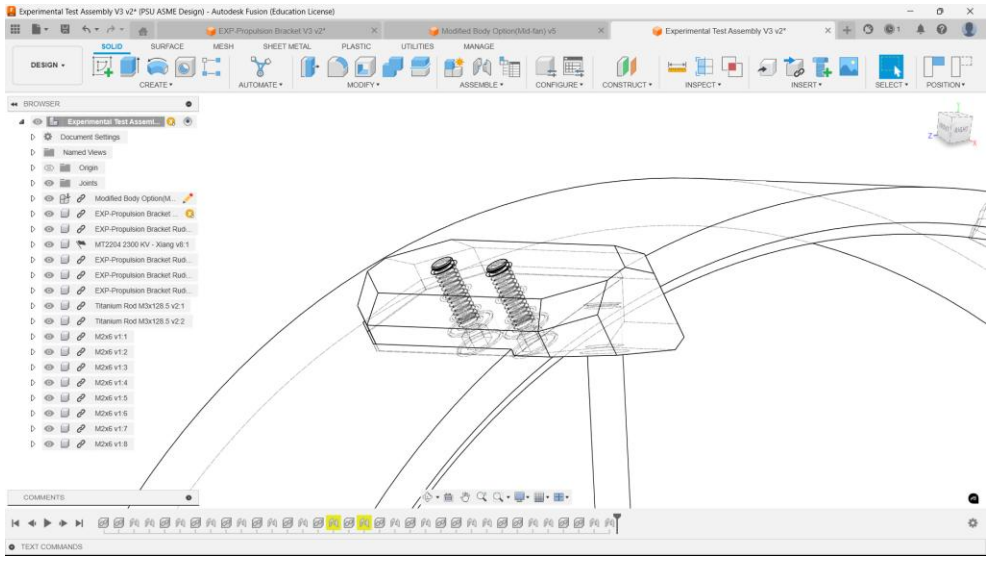


Rudder Rework!

Previously the rudders pins were plastic. They keep breaking

Do we want to change to metal?





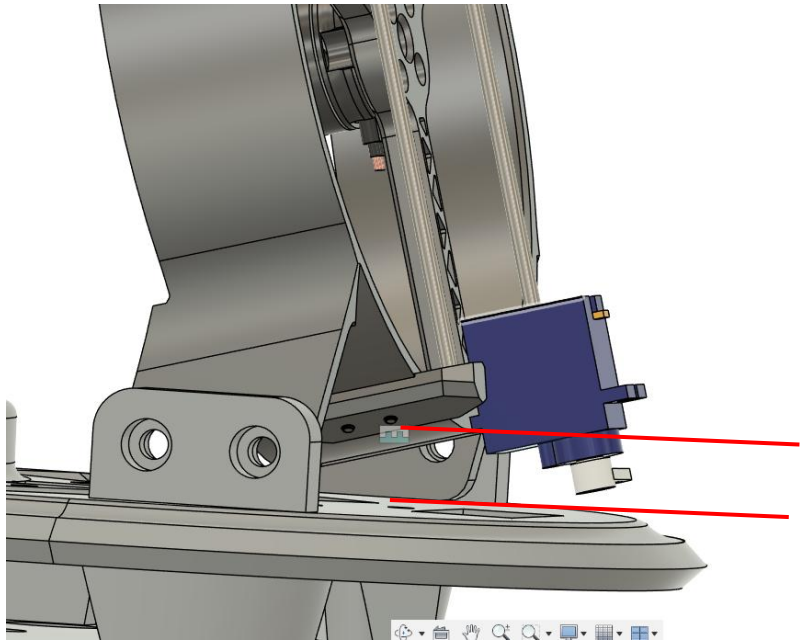
Rudder Rework!



Check this out we got titanium rods

They were only like 7 dollars too

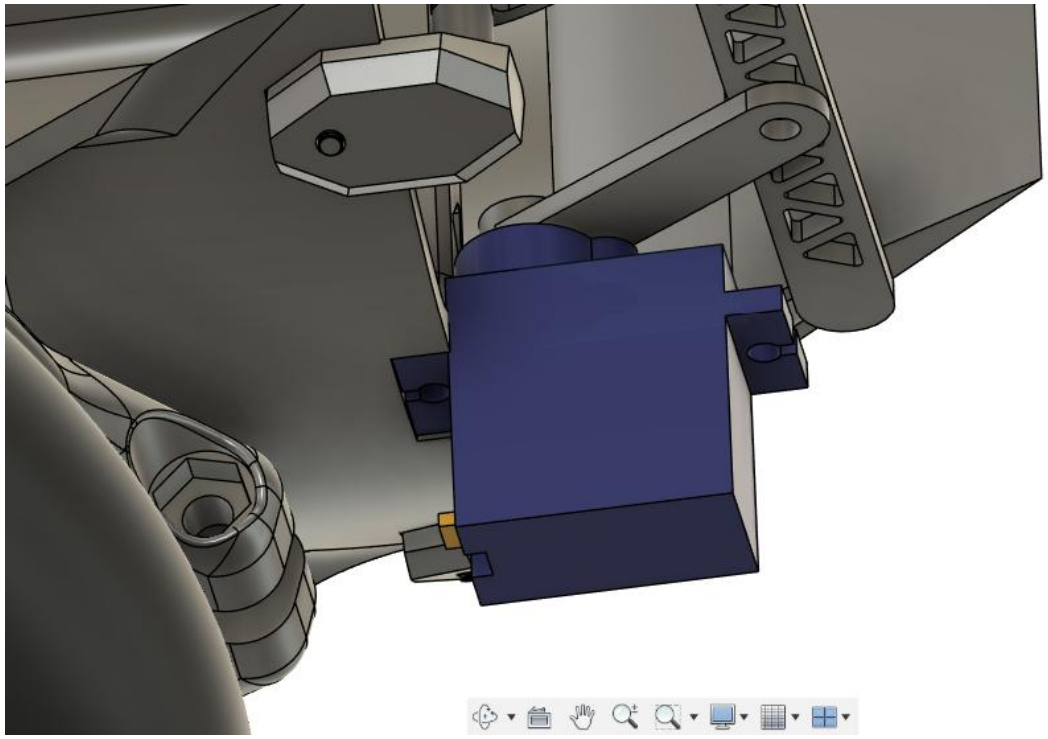
Thoughts on mounting the servo upside down?

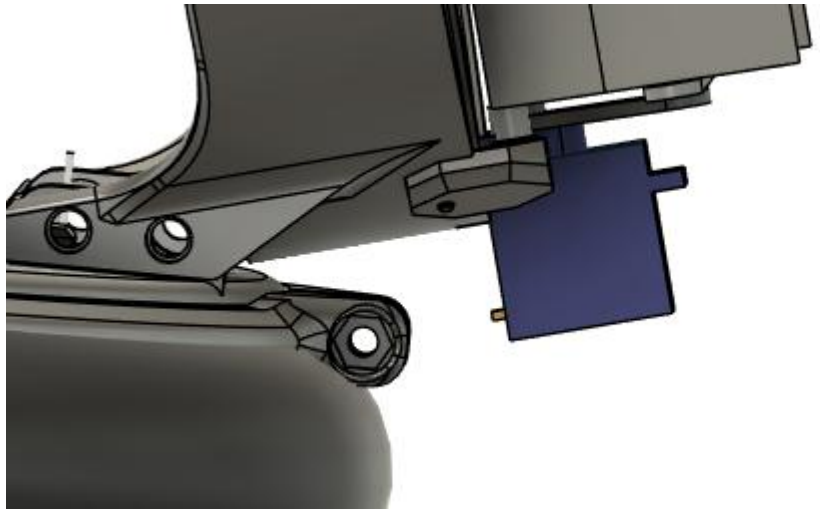


Might help the clearance here

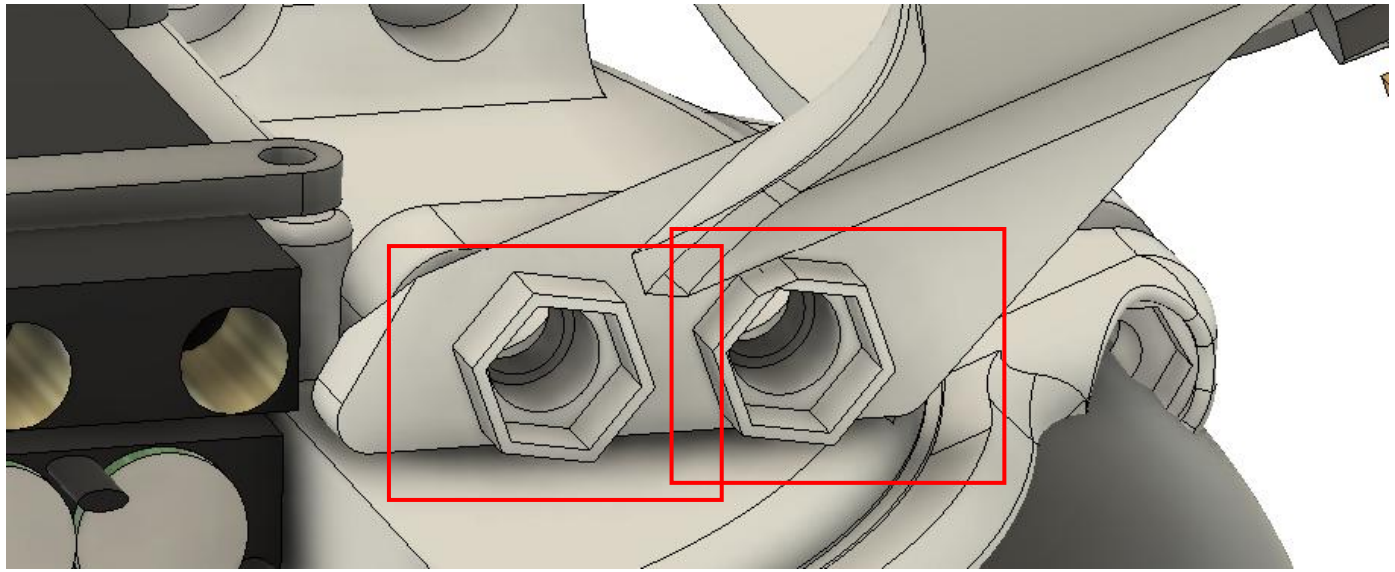
Maybe not. It makes the wiring too far. It also stands very far away.

We can extend it over the edge instead but right side up





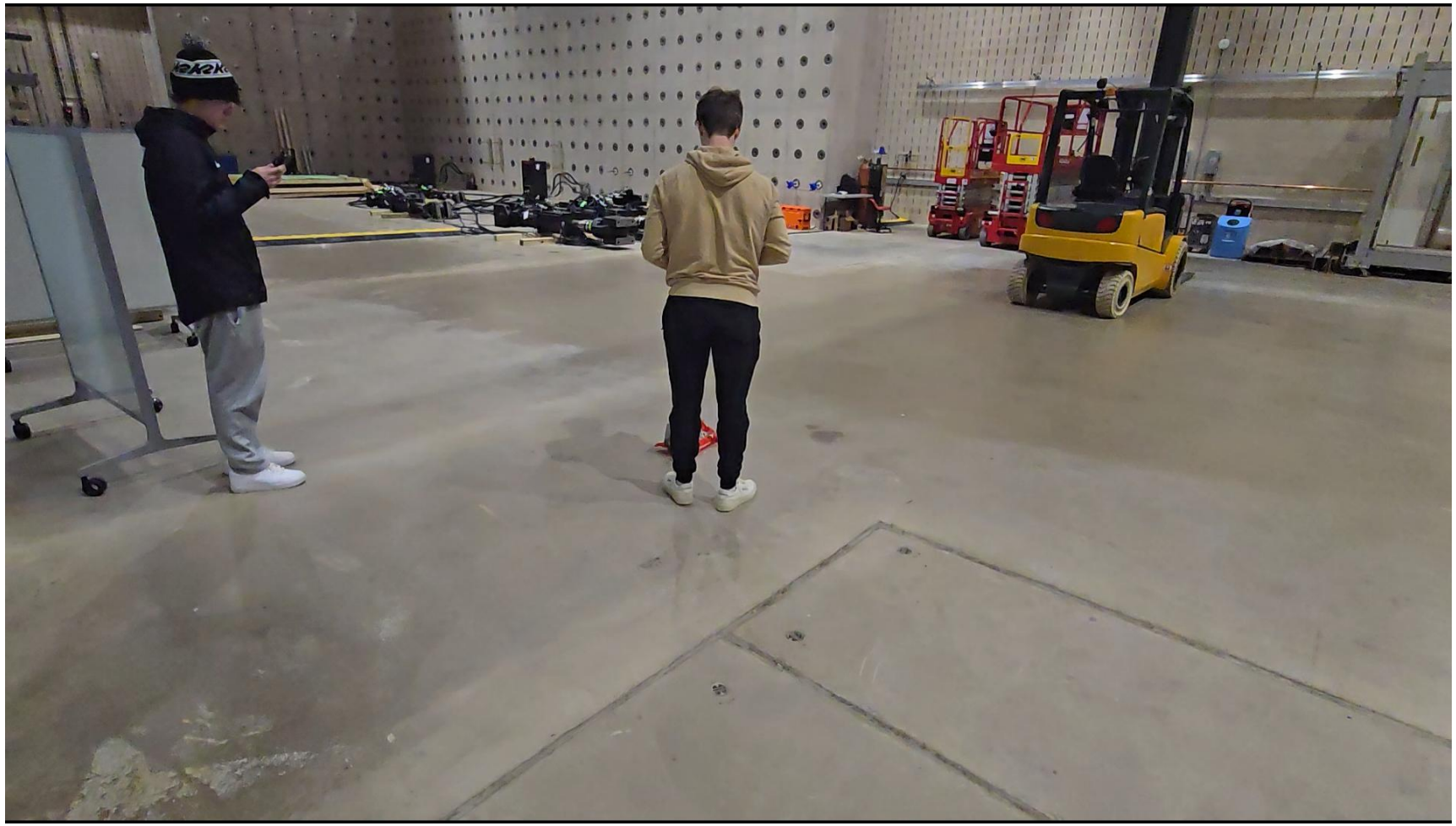
I don’t think this is necessary



Definitely helps with nut staying in place.

But friction is good enough

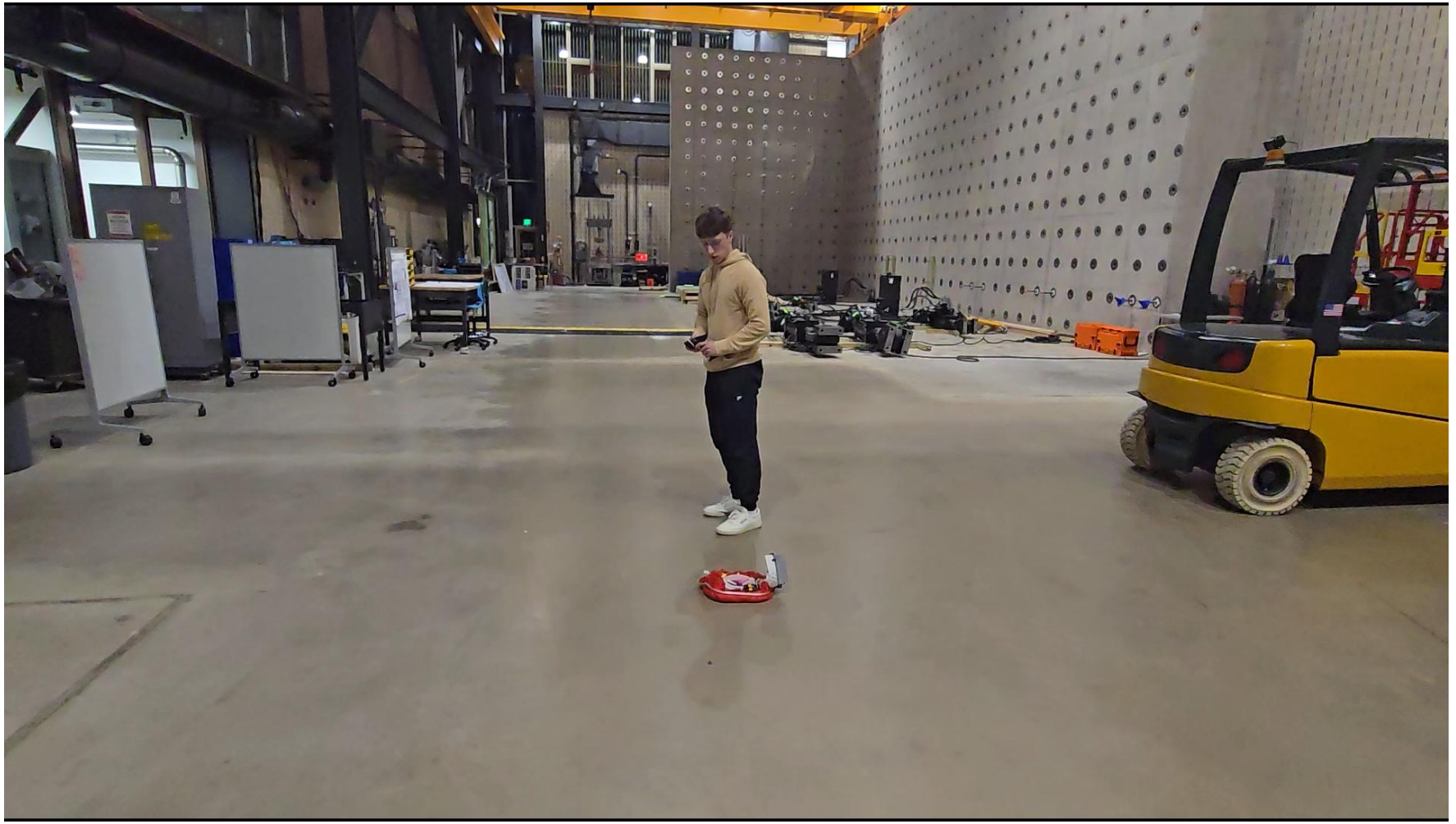
Testing 2! First video



Testing maneuverability

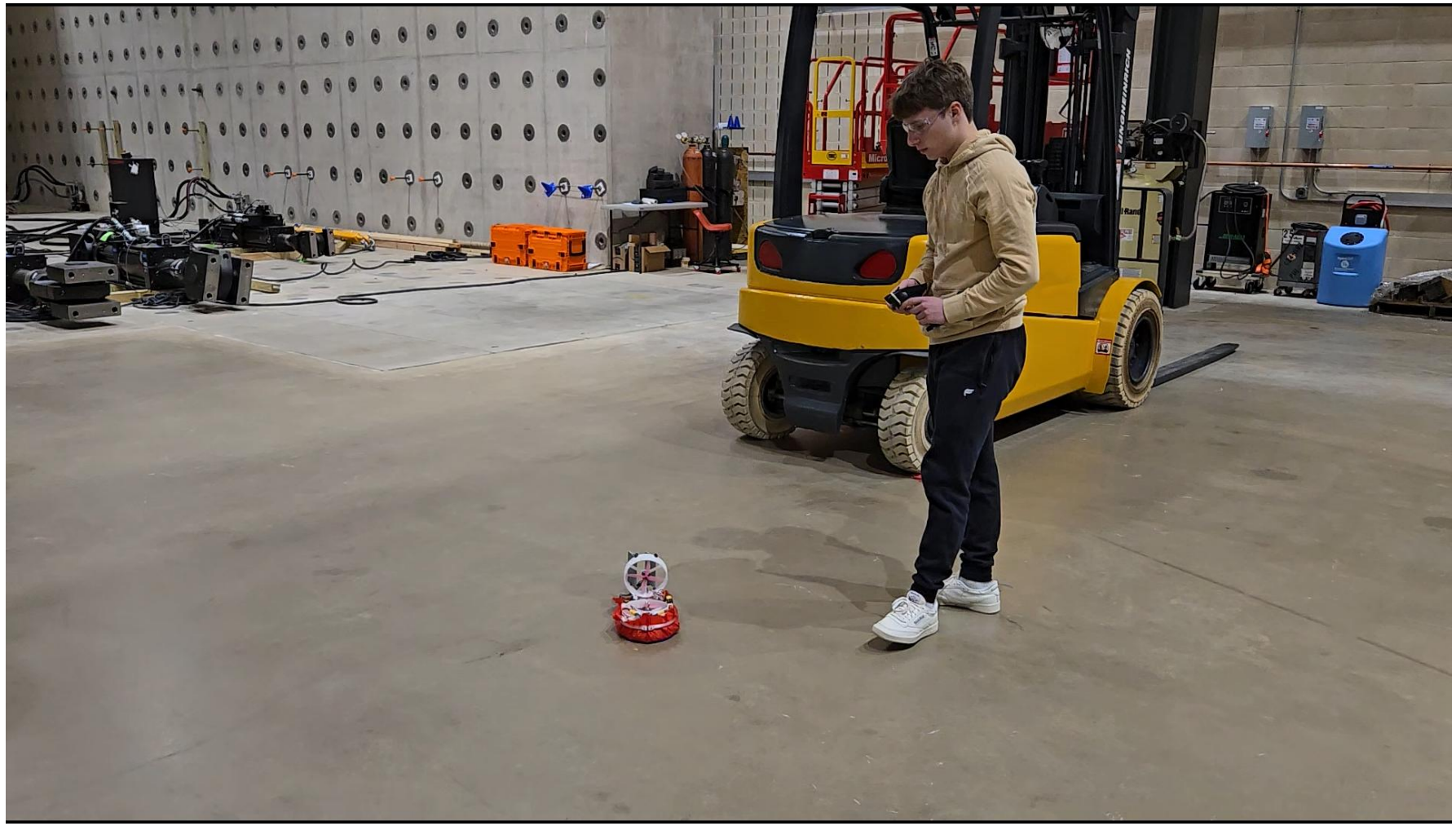
Response time

Testing 2! Second video



Excess skirt material getting sucked into lift fan

Testing 2! Third Video

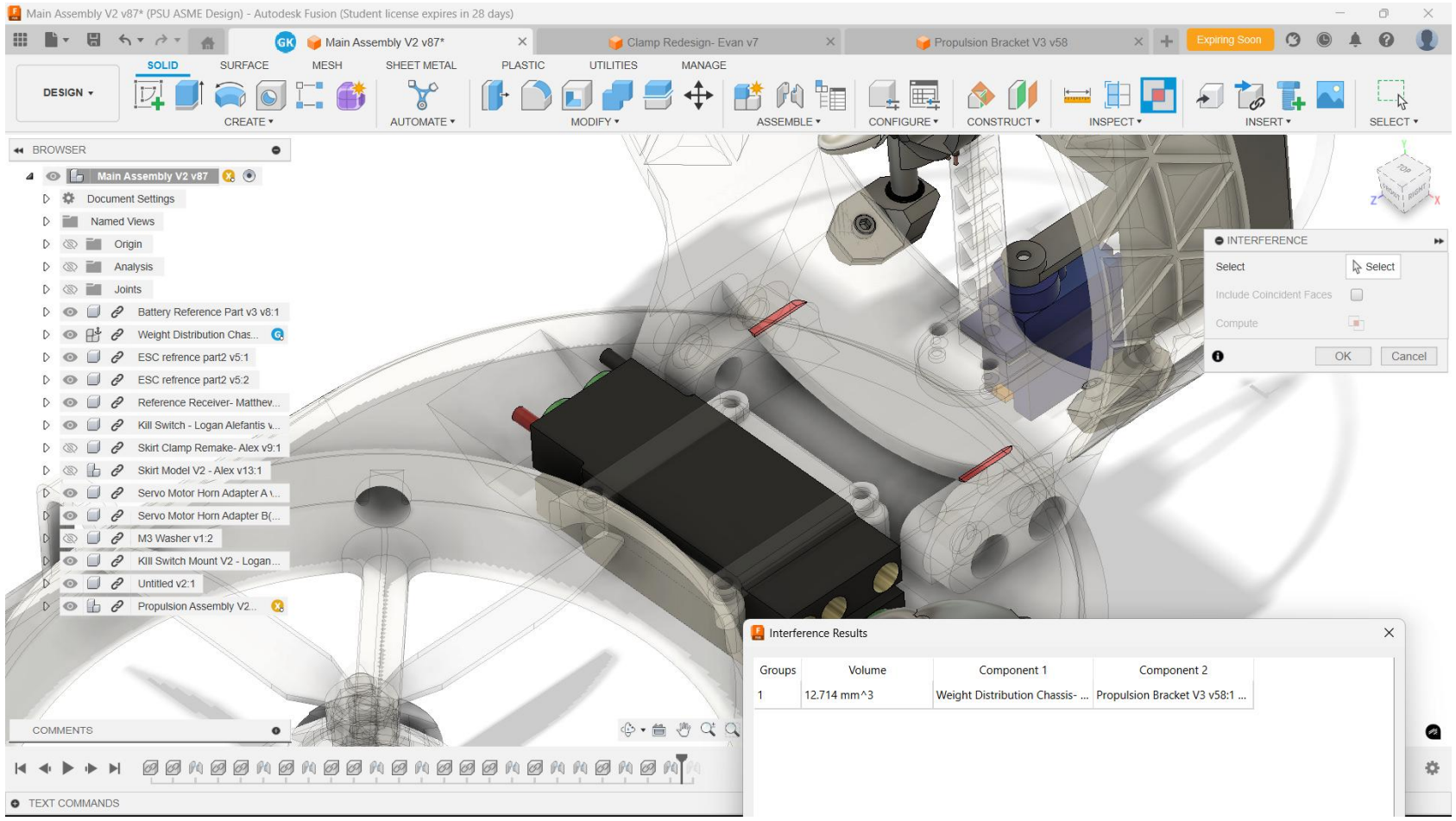


Precision testing

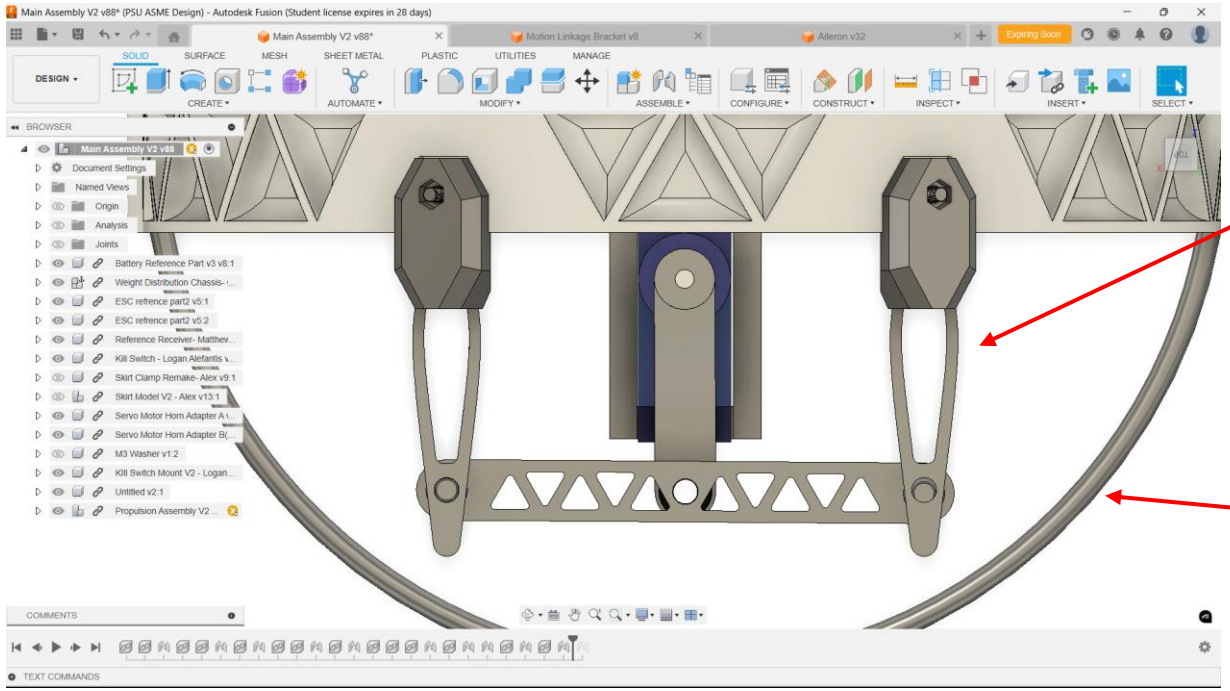
oops



Interference



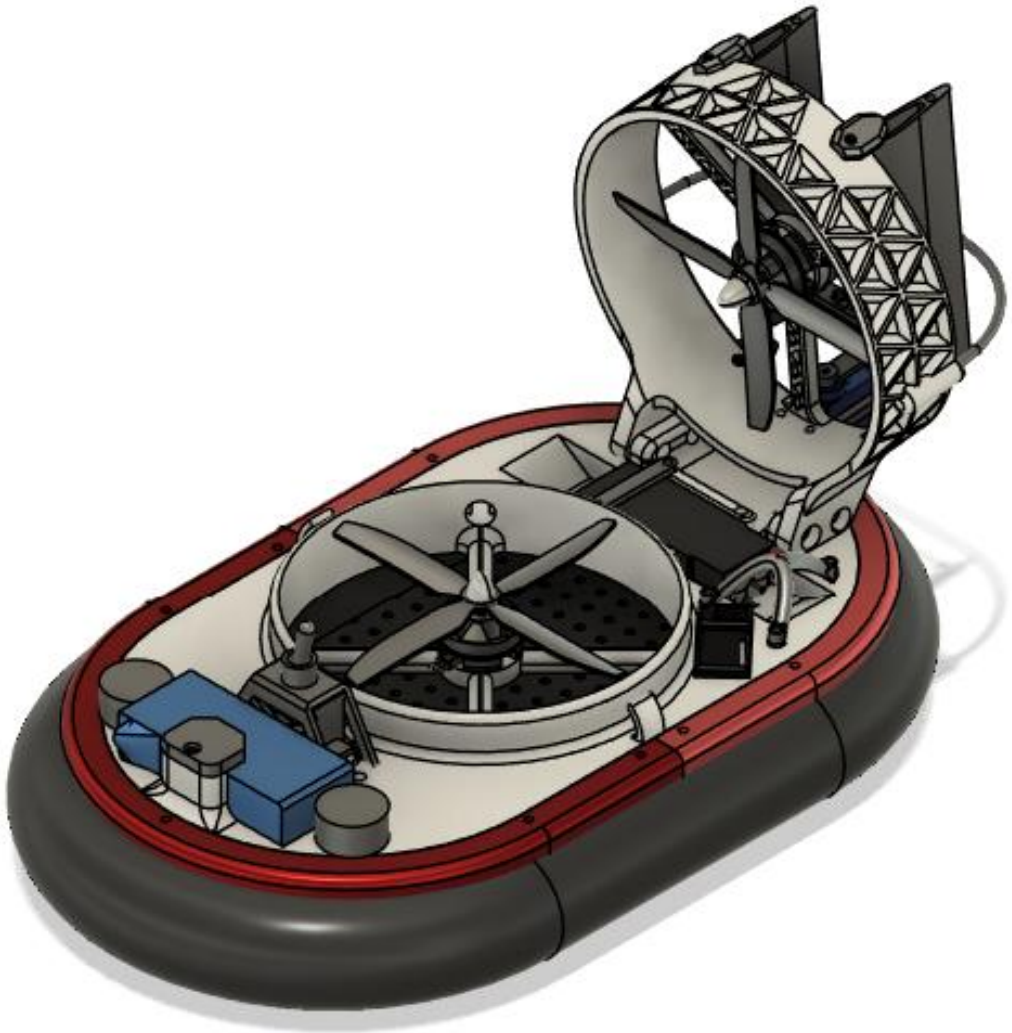
Changes



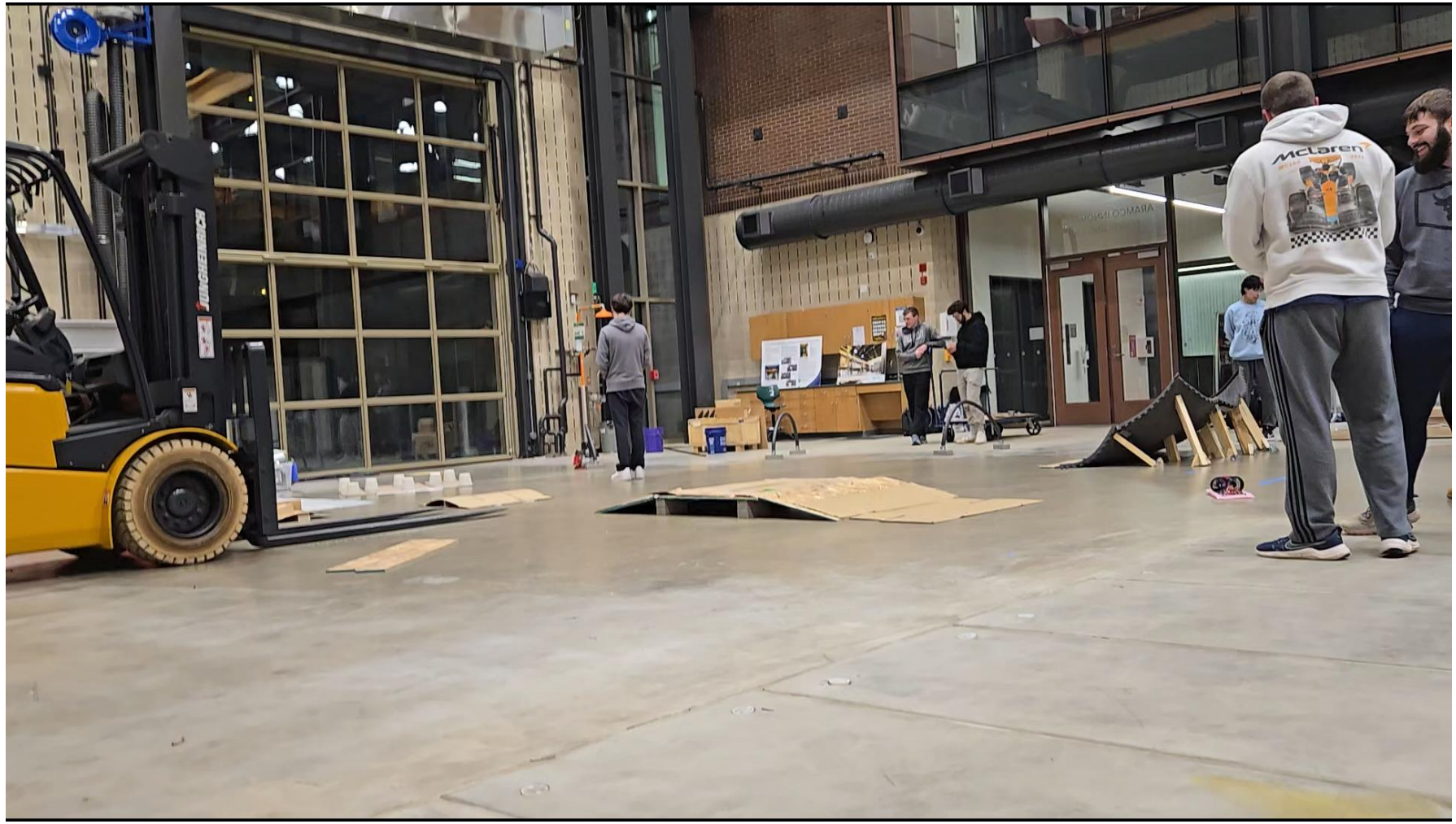
Made this profile more efficient in forward flight

Added Protective Halo

Final Design

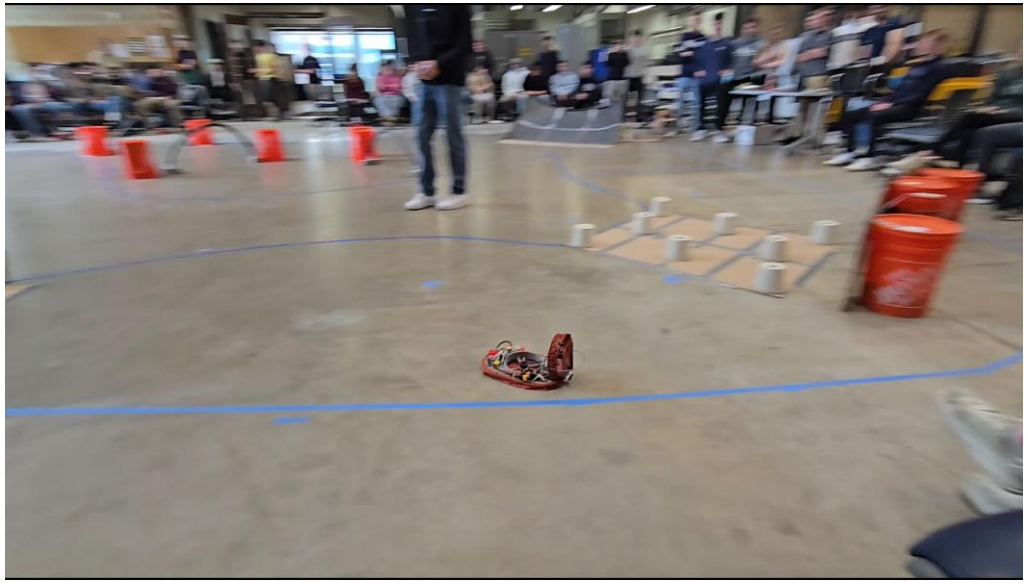


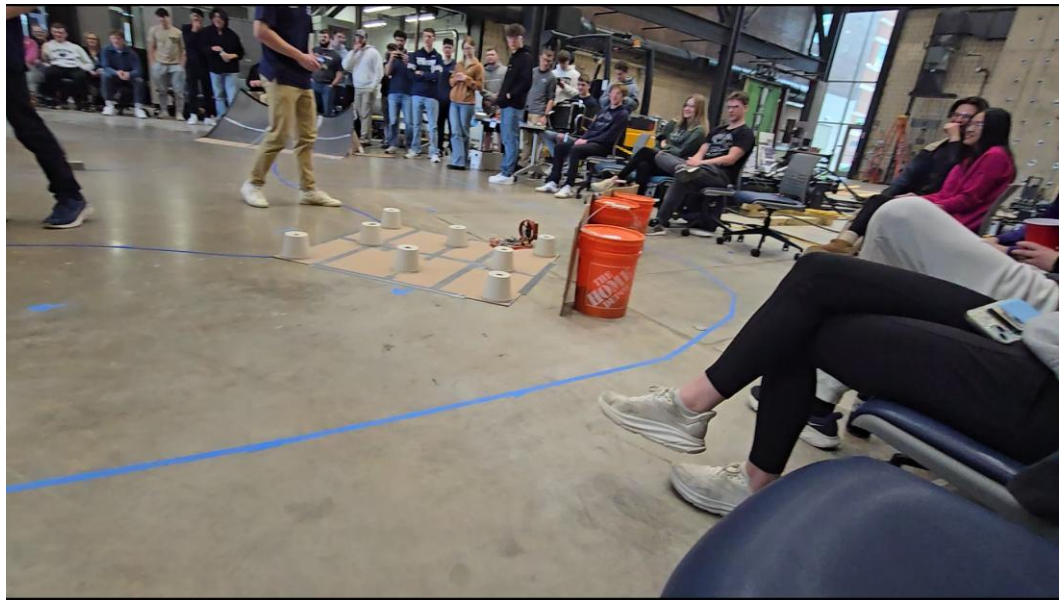
Practice before comp day!



Comp Day Footage!





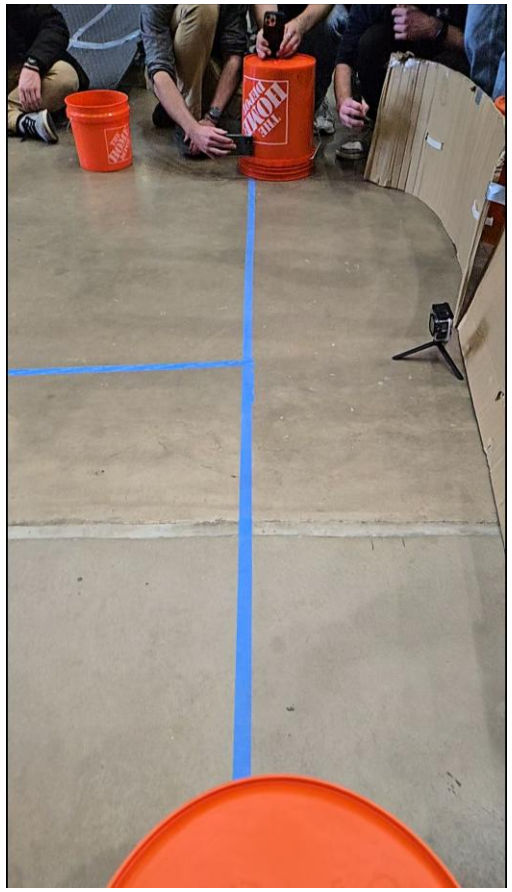


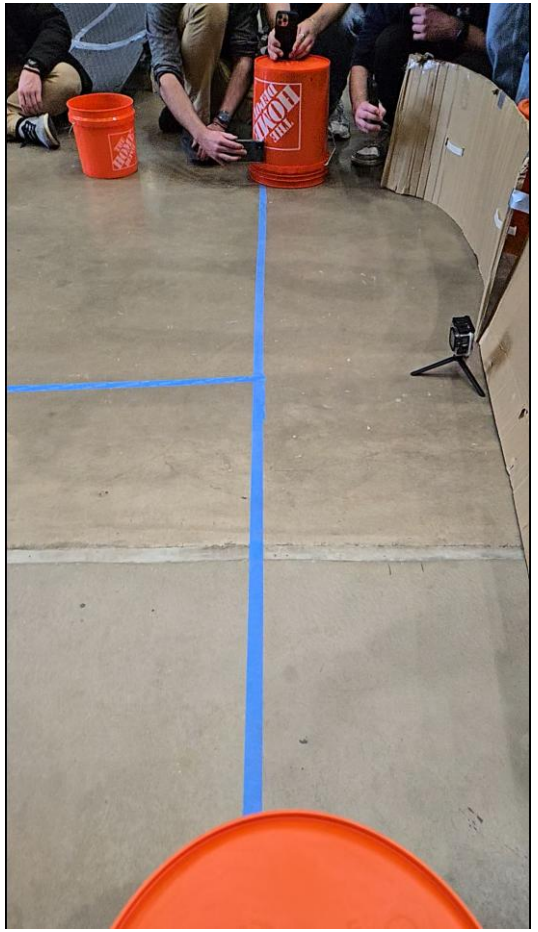
Drag Race 1



Spoiler… We ended up tieing so we had to redo it.

Drag Race 2





Looking Good!



The team!

That's me!

